PRODUCT USER MANUAL

For Ocean Colour Products

OCEANCOLOUR_GLO_BGC_L3_NRT_009_101
OCEANCOLOUR_GLO_BGC_L4_NRT_009_102
OCEANCOLOUR_GLO_BGC_L3_MY_009_103
OCEANCOLOUR_GLO_BGC_L4_MY_009_104
OCEANCOLOUR_GLO_BGC_L3_MY_009_107
OCEANCOLOUR_GLO_BGC_L4_MY_009_108
OCEANCOLOUR_ATL_BGC_L3_NRT_009_111
OCEANCOLOUR_ATL_BGC_L4_NRT_009_112
OCEANCOLOUR_ATL_BGC_L3_MY_009_113
OCEANCOLOUR_ATL_BGC_L4_MY_009_114
OCEANCOLOUR_ATL_BGC_L4_NRT_009_116
OCEANCOLOUR_ATL_BGC_L4_MY_009_118
OCEANCOLOUR_ARC_BGC_L3_NRT_009_121
OCEANCOLOUR_ARC_BGC_L4_NRT_009_122
OCEANCOLOUR_ARC_BGC_L3_NRT_009_123
OCEANCOLOUR_ARC_BGC_L4_NRT_009_124
OCEANCOLOUR_BAL_BGC_L3_NRT_009_131
OCEANCOLOUR_BAL_BGC_L4_NRT_009_132
OCEANCOLOUR_BAL_BGC_L3_MY_009_133
OCEANCOLOUR_MED_BGC_L3_NRT_009_141
OCEANCOLOUR_MED_BGC_L4_NRT_009_142
OCEANCOLOUR_MED_BGC_L3_MY_009_143
OCEANCOLOUR_MED_BGC_L4_MY_009_144
OCEANCOLOUR_BLK_BGC_L3_NRT_009_151
OCEANCOLOUR_BLK_BGC_L4_NRT_009_152
OCEANCOLOUR_BLK_BGC_L3_MY_009_153
OCEANCOLOUR_BLK_BGC_L4_MY_009_154

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RECORD TABLE

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7.0	2025/11/12	All	Updates for March 2026 EIS	S. Colella, E. Böhm, C. Cesarini, Q. Jutard	V. Brando

Table of Contents

GLOSSARY AND ABBREVIATIONS *5*

DATA ACCESS **6**

I INTRODUCTION *7*

I.1 Summary **7**

I.2 History of changes **11**

II DESCRIPTION OF THE PRODUCT SPECIFICATION *19*

II.1 General Information **19**

II.1.1 Product Levels 19

II.1.2 Product processing mode and temporal coverage 19

II.1.3 Operational Near real Time (NRT) products 20

II.1.4 Reprocessed (MY) products 20

II.2 Details of the products **21**

II.2.1 Global Ocean (Copernicus-GlobColour) 22

II.2.2 Global Ocean (OC-CCI) 24

II.2.3 Atlantic 25

II.2.4 Artic Region 28

II.2.5 Baltic Sea 29

II.2.6 Mediterranean Sea 31

II.2.7 Black Sea 33

II.3 Details of datasets **35**

II.3.1 Global Ocean (Copernicus-GlobColour) 35

II.3.2 Global Ocean (OC-CCI) 62

II.3.3 Atlantic 64

II.3.4 Arctic Region 79

II.3.5 Baltic Sea 85

II.3.6 Mediterranean Sea 96

II.3.7 Black Sea 110

II.4 Additional Variables **121**

II.4.1 Mediterranean Sea, Black Sea and Baltic Sea Sensor additional masks 121

II.4.2 Mediterranean Sea, Black Sea, Baltic Sea, Atlantic and Arctic Quality Index 121

II.4.3 Global Ocean and Atlantic Quality Flag and Uncertainties (Copernicus-GlobColour products) 122

II.5 Upstream Data Unavailability **122**

III FILE FORMAT 124

IV FILES NOMENCLATURE 125

IV.1 Nomenclature of files when downloaded through the Copernicus Marine Web Portal Subset Service 125

IV.2 Nomenclature of original files 127

IV.3 Domain Coverage 128

IV.4 Structure of files 130

V DIFFERENCE BETWEEN COPERNICUS-GLOBCOLOUR AND OC-CCI IN GLO MY PRODUCTS 131

VI REFERENCES 136

GLOSSARY AND ABBREVIATIONS

PC	Production Centre
PU	Production Unit
OCTAC	Ocean Colour Thematic Assembly Centre
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
ARC	Arctic Sea
ATL	Atlantic Sea
BAL	Baltic Sea
BLK	Black Sea
GLO	Global Ocean
MED	Mediterranean Sea
CNR	Italian National Research Council
ACRI	ACRI-ST
PML	Plymouth Marine Laboratory
BC	Brockmann Consult
PP	Primary Production
PUM	Product User Manual
CHL	mass concentration of chlorophyll_a in seawater
CDM	volume absorption coefficient of radiative flux in seawater due to dissolved organic matter and non algal particles
K490	volume attenuation coefficient at 490nm of downwelling radiative flux in seawater
RRS	surface ratio of upwelling radiance emerging from seawater to downwelling radiative flux in air
BBP443	volume backwards scattering coefficient at 443nm of radiative flux in seawater due to particles
ZSD	Secchi depth of seawater
IOPs	Inherent Optical Properties

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1 Summary

The OC Production Centre implemented a European Ocean Colour Service for Copernicus marine applications providing global, European and regional (Atlantic, Arctic, Baltic, Mediterranean, and Black Seas) high quality core OC products. OCTAC products are currently used by European, National and regional level intermediate/end users (intergovernmental bodies, National Environmental Agencies, ESA and European Projects etc.) to monitor marine conditions and are acquired by the Copernicus Marine Service MFCs for ecosystem model assimilation/validation.

In the OC Production Centre (PC) there are 3 Production Units (PU) CNR, ACRI, PML. Presently functions assigned amongst the institutions are:

- SS Service Desk for OCTAC and single point of contact for communication of incidents changes and service requests by users (CNR)
- Production Units:
    - ACRI: operating GLO and ATL (L4-interpolated) NRT production chains and providing corresponding MY (based on the Copernicus-GlobColour processor).
    - CNR: operating the MED, BLK and BAL and ARC production chains in NRT and MY mode
    - BC: providing GLO MY products developed in the framework of ESA OC-CCI/C3S Project (taken over from PML)

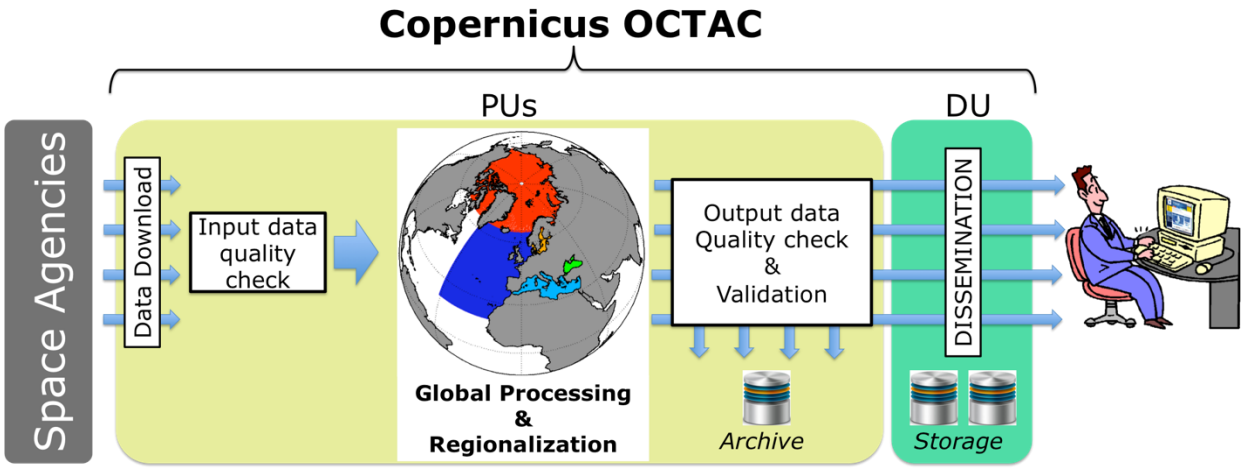


Figure 1: Overview of the OC TAC system

At each PU, the processing sequences can be divided into 5 main functions:

1. Data Ingestion

2. Products Generation
3. Products conversion and storage
4. Quality control
5. System Monitoring

The OCTAC Ocean regions can be therefore: GLO (global ocean), ARC (Arctic Ocean), BAL (Baltic Sea), ATL (Atlantic Ocean), MED (Mediterranean Sea), and BS (Black Sea). Table 1 below shows the geographical limits for each region.

Area Code	Ocean Region	Southernmost Latitude	Northernmost Latitude	Westernmost Longitude	Easternmost Longitude
GLO	Global Ocean	90.0°S	90.0°N	180.0°W	180.0°E
ARC	Arctic Ocean	65.0°N	90.0°N	180.0°W	180.0°E
BAL	Baltic Sea	53.25°N	65.85°N	9.25°E	30.25°E
ATL	Atlantic Ocean	20.0°N	66.0°N	46.0°W	13.0°E
MED	Mediterranean Sea	30.0°N	46.0°N	6.0°W	36.5°E
BLK	Black Sea	40.0°N	48.0°N	26.5°E	42.0°E

Table 1: Geographical limits for Copernicus Marine Service OCTAC ocean regions.

For each region, OCTAC delivers only one type of product containing up to five datasets: i) **plankton** with the phytoplankton chlorophyll concentration evaluated either via standard processing or via region-specific algorithms and Phytoplankton Functional Types (PFTs); ii) **Primary Production** (PP); iii) **reflectance** with the spectral Remote Sensing Reflectance (Rrs); iv) **transparency**; with diffuse attenuation coefficient of light at 490 nm (Kd490), Secchi depth (transparency of water), and the non-organic Solid Particulate Matter (SPM); and v) **optics** including the IOPs (Inherent Optical Properties) such as absorpti **mems_obs-oc_blk_bgc-pp_nrt_l4-multi-4k** on and scattering and particulate and dissolved matter.

The **product level** can be Level-3 (**L3**) or Level-4 (**L4**). L3 are the daily composite products as obtained by merging all the ocean satellite passages and they can be of any spatial resolution, typically 1, 4, 9, and 25 km. L4 are those products for which a temporal averaging method or an interpolation procedure is applied to fill in missing data values. Temporal averaging is performed on a monthly basis. The interpolation procedure currently used spans from Optimal Interpolation to DINEOF procedure. More details about product levels are provided in section II.1.1.

Following the Copernicus Marine Service convention, the OC products are further classified in **NRT (Near Real Time)** and **MY (Multi-Year/Reprocessed)**. More details about product processing mode and temporal coverage are provided in section II.1.2.

The **NRT products** are operationally produced every day and provide the best estimate of the ocean colour variables at the time of processing. A first version of this product is generated soon after the satellite passage, by using climatological ancillary data (meteorological and ozone data for atmospheric correction, and attitude and ephemerides for data geolocation). These versions of NRT products are available within 24 hours from the satellite acquisition. The

NRT products are then updated by reprocessing the OC data by using updated ancillary information. The new files (defined MYINT) are available within 31 days from sensing (though typically around 15-20, depending from the input data and upstream providers). The consistency of the OC NRT product time series is monitored by the OCTAC production units. Info about the product quality is detailed in relevant QUality Information Documents, QUIDs.

The **MY** products, instead, are consistent multi-year time series produced by using a consolidated and consistent input dataset, with a unique processing software configuration. Therefore, they represent a much more solid data set for long-term analyses. New to this version, multi year interim production will be carried out to extend the multi-year time series using the same configuration of the previous full reprocessing. More details about product versions are provided in section II.1.2. It should be noted that some daily products are missing during the period [1997-2002] due to the non-availability of the SeaWIFS input data, which was the only sensor operational in that period.

For NRT products, presently the entire OCTAC relies on the available OC sensors: MODIS-Aqua, NPP-VIIRS and NOAA20-VIIRS (from NASA) and Sentinel-3 OLCI A and B (from EUMETSAT). The data from Sentinel-2 A & B MSI (from ESA) is used for the High resolution products, that are described separately in OCEANCOLOUR_HR Product User Manual (CMEMS-HR-OC-PUM-009-201to212_v1.0). ACRI PU products (GLO and ATL) also use NOAA21-VIIRS and has retired MODIS-Aqua from real-time production.

It should be noted that these sensors are able to retrieve data by measuring the sun reflection over the sea. It means during the night or with cloudy conditions no data are available. As a consequence, at high latitude it is normal that products over Baltic or Arctic are not available or with a limited coverage during winter.

Products available from OCTAC are listed below:

- Global (Copernicus-GlobColour)
  - OCEANCOLOUR_GLO_BGC_L3_NRT_009_101
  - OCEANCOLOUR_GLO_BGC_L4_NRT_009_102
  - OCEANCOLOUR_GLO_BGC_L3_MY_009_103
  - OCEANCOLOUR_GLO_BGC_L4_MY_009_104
- Global (OC-CCI/C3S)
  - OCEANCOLOUR_GLO_BGC_L3_MY_009_107
  - OCEANCOLOUR_GLO_BGC_L4_MY_009_108
- Atlantic
  - OCEANCOLOUR_ATL_BGC_L3_NRT_009_111
  - OCEANCOLOUR_ATL_BGC_L3_MY_009_113
  - OCEANCOLOUR_ATL_BGC_L4_NRT_009_116
  - OCEANCOLOUR_ATL_BGC_L4_MY_009_118
- Arctic
  - OCEANCOLOUR_ARC_BGC_L3_NRT_009_121

    - OCEANCOLOUR_ARC_BGC_L4_NRT_009_122
    - OCEANCOLOUR_ARC_BGC_L3_NRT_009_123
    - OCEANCOLOUR_ARC_BGC_L4_NRT_009_124
- Baltic
    - OCEANCOLOUR_BAL_BGC_L3_NRT_009_131
    - OCEANCOLOUR_BAL_BGC_L4_NRT_009_132
    - OCEANCOLOUR_BAL_BGC_L3_MY_009_133
    - OCEANCOLOUR_BAL_BGC_L4_MY_009_134
- Mediterranean Sea
    - OCEANCOLOUR_MED_BGC_L3_NRT_009_141
    - OCEANCOLOUR_MED_BGC_L4_NRT_009_142
    - OCEANCOLOUR_MED_BGC_L3_MY_009_143
    - OCEANCOLOUR_MED_BGC_L4_MY_009_144
- Black Sea
    - OCEANCOLOUR_BLK_BGC_L3_NRT_009_151
    - OCEANCOLOUR_BLK_BGC_L4_NRT_009_152
    - OCEANCOLOUR_BLK_BGC_L3_MY_009_153
    - OCEANCOLOUR_BLK_BGC_L4_MY_009_154

I.2 History of changes

This section describes the updates of the OCTAC system occurred since May 2015.

Date	Description of changes and impacted product
04/2016	**ACRI** ▪ The Copernicus-GlobColour REP time-series [1997-2014] is now provided. Weaknesses and strengths of Copernicus-GlobColour compared to OC-CCI are provided in section IV ▪ The NRT datasets start in 2015, ▪ 8-days and monthly products are now available, ▪ A non-mineral SPM product has been added. ▪ Global Optics products at 25 and 100 km have been retired ▪ BAL NRT production moved from ACRI PU to CNR PU ▪ Baltic GSM chlorophyll dataset [2012-2014] have been retired (superseded by a Neural Network approach, see CNR below) **CNR** ▪ Only one chlorophyll product is now produced using different algorithm for open and coastal waters. This is labelled “CHL” instead of being labelled case 1, case 2. ▪ An L4 NRT product is now provided for the MED and BS areas (both optics and chlorophyll). This is provided as 8-day and monthly mean composite products from MODIS and VIIRS ▪ In addition to daily L4 CHL REP product, an L4 CHL REP product is now provided for the MED and BS areas. This is provided as 8-day and monthly mean composites based on the L3 REP. ▪ The BAL CHL L3 REP is now based on CCI reflectance and using a new regional algorithm. **PML** ▪ Only one chlorophyll product is now produced (OC5CI). This is labelled “chl” instead of being labelled with the algorithm name. ▪ An L4 NRT product is now provided for the ARC and ATL areas (both optics and chlorophyll). This is provided as 8-day and monthly mean composite products from MODIS and VIIRS ▪ An L4 CHL REP product is now provided for the ARC and ATL areas. This is provided as 8-day and monthly mean composites based on the L3 REP. ▪ An L4 GLO REP CHL product is now provided based on OC-CCI data, provided as 8-day and monthly mean composites. ▪ Temporal coverage of the L3 GLO REP CHL (4km) product from OC-CCI has been extended to 31st December 2014 and rebased on the newer OC-CCI v2.0 release. The coverage of the regional L3 GLO REP (1km) is similarly extended.

	▪ Chlorophyll products OC5 and OC488 have been superseded by OC5CI and retired
06/2016	**ACRI/CNR** ▪ BAL NRT production moved from ACRI PU to CNR PU ▪ Baltic OPT NRT ACRI multi-sensor datasets have been retired. ▪ New BAL OPT NRT datasets single-sensor (modis_a) released.
07/2016	**ACRI** ▪ The daily L4 GLO interpolated (cloud free) product is now available as a REP [1997-2014] OCEANCOLOUR_GLO_CHL_L4_REP_OBSERVATIONS_009_082
09/2016	**ACRI** ▪ The daily L4 ATL interpolated (cloud free) product is now available as a REP [1997-2014] OCEANCOLOUR_ATL_CHL_L4_REP_OBSERVATIONS_009_098
04/2017	**ACRI** ▪ The Copernicus-GlobColour REP time-series have been extended to [1997-Aug-2016] using as inputs the last official level2 products from agencies. ▪ The NRT datasets start in Sep-2016 **CNR** ▪ Level-2 replaced Level-1 as upstream data for all the available sensors. This enables this production chain to more efficiently and promptly react and incorporate changes in space agencies' production chains; ▪ Updated destriping procedure ▪ Updated bowtie procedure ▪ Multi-sensor products replaced single-sensor products, which are no longer distributed; ▪ Updated of all chlorophyll algorithms ▪ Updated of Kd algorithm for Mediterranean Sea ▪ All REP product updated with new version of CCI processor (v3) ▪ BAL REP computed at 1 km of resolution ▪ Addition of new datasets within the Optics category: Inherent Optical Properties such as the phytoplankton absorption coefficient, the detritus absorption coefficient, and the particulate backscattering coefficient, all three at 443 nm ▪ REP Level-4 products only distributed at monthly or 8-day temporal resolution. **PML** ▪ New multi-sensor merged Chlorophyll and RRS products for ATL and ARC areas in DT are now being produced, essentially identical to those provided as REP through OC-CCI

	▪ Input data stream switched from L1 to L2, where possible (not all products can be provided from L2 inputs), for consistency across PUs ▪ Temporal coverage of the L3 GLO REP CHL (4km) product from OC-CCI has been extended to 31st December 2015 and rebased on the newer OC-CCI v3.0 release. The coverage of the regional L3 GLO REP (1km) is similarly extended.
10/2017	**ACRI** ▪ The OLCI single-sensor products are now available for the period [6-July-2017, present]. The full mission (since April 2016) will be completed when it will have been reprocessed by ESA. **CNR** ▪ The OLCI single-sensor products are now available for the period [6-July-2017, present]. The full mission (since April 2016) will be completed when it will have been reprocessed by ESA. **PML** ▪ The OLCI single-sensor products are now available for the period [6-July-2017, present]. The full mission (since April 2016) will be completed when it will have been reprocessed by ESA. ▪ The multi-sensor CCI product has been brought up to full NRT status. Both NRT and DT versions of this product are now produced.
12/2017	**All Pus** ▪ Temporal coverage of all multi-sensor REP products has been extended to the end of 2016 **ACRI** ▪ A GSM chlorophyll climatology is now provided at Global level.
04/2018	**All Pus** ▪ the NRT MODIS and VIIRS datasets are using the R2018.0 NASA products since December 2017. It is planed to update all the CMEMS products using this new data in the coming months. **ACRI** ▪ Temporal coverage of the Global NRT OLCI product has been extended to start the 25th of April 2016 using the last baseline (2.23). **PML** ▪ Single-sensor MODIS and VIIRS products have been withdrawn from the catalogue, with the exception of MODIS ADG, APH and ATOT and VIIRS SPM. The withdrawn products have been superseded by single-sensor OLCI products and by multi-sensor products based on OC-CCI, which are now available in NRT.

	▪ Dimensions of the ATL and ARC areas have changed slightly. This reflects an upgrade of mapping software at PML PU, which now uses a common grid for all output products.
06/2018	**ACRI** ▪ The "cloud free" (interpolated) chlorophyll products (NRT & REP) have been updated to take benefit of the last R2018 NASA reprocessing and use of OLCI-S3A. **PML** ▪ Temporal coverage of the Global NRT OLCI product has been extended to start the 25th of April 2016 using the last baseline (2.23).
11/2018	**ACRI** ▪ Treatment of oligotrophic waters has been improved for both NRT and REP datasets **PML** ▪ Change of chlorophyll algorithm from OC5C to OC5 for ATL and ARC **CNR** ▪ Update of multi sensor processor for NRT time series for MED and BS
04/2019	**CNR** ▪ Update of chlorophyll algorithm for BS (NRT/DT and REP) ▪ Addition of Phytoplankton Functional Types expressed as chlorophyll_a in seawater for MED within the CHL product
12/2019	**ACRI** ▪ Use of OLCI-S3B and VIIRS-SNPP (NOAA20) for Chlorophyll product, ▪ Addition of the "Primary Production" products (NRT and REP), ▪ Addition of a "KD" cloud free product (NRT and REP), ▪ Names of the chlorophyll datasets are changed, ▪ Sensor drifts better corrected (consistency of the long time series [1997-present]). ▪ Retirements of 8-days, 25 and 100km datasets. **PML** ▪ ARC, ATL and GLO NRT/DT and REP retirement of 8-days composites CHL and OPTICS ▪ ARC, ATL and GLO, REP - update with new version of CCI processor (v4) and temporal extensions (this includes NASA R2018 updates to input data and extension of time series to end of 2018). ▪ ATL and GLO, REP – introducing PFT dataset with PSC variables in CHL product

	CNR ▪ MED and BS, NRT/DT and REP - Retirement of 8-day composites CHL and OPTICS ▪ MED and BS REP - update with new version of CCI processor (v4) and temporal extension; ▪ MED REP - Introducing PSC variables in PFT dataset in CHL product, and temporal extension; ▪ MED and BS NRT/DT – inclusion of OLCI in multi-sensor products; ▪ MED and BS NRT/DT - OLCI single-sensor products are updated with the BRDF correction and improved CHL algorithms; ▪ BAL NRT - Retirement of MODIS single sensor products; ▪ BAL NRT/DT - OLCI single sensor: new processing with BALHZG4OLCI_v2; ▪ BAL REP - update with new version of CCI processor (v4), new bio-optical characterization (rrs bias adjustment with Aeronet-OC data and new chlorophyll algorithm) and temporal extension;
04/2020	**PML** ▪ ATL and GLO, REP – introducing PFT dataset with PSC variables in CHL product ▪ ARC and ATL, NRT/DT – introducing QI variable in OLCI sensor datasets ▪ ARC and ATL, NRT/DT – update with new version of CCI processor (v4) ▪ ARC and ATL, NRT/DT multi datasets temporal extension
07/2020	**ACRI** ▪ GLO: new PFT and interpolated ZSD datasets, updated PP with better coastal spatial resolution and temporal extension. ▪ ATL: temporal extension **CNR** ▪ BAL: 3 new IOP datasets introduced: bbp510, aph440, adg440 ▪ REP temporal extension for MED, BS and BAL **PML** ▪ Change of chlorophyll algorithm from OC5CI to OC5CCI for ARC and ATL NRT/DT and REP ▪ Temporal extension for ARC, ATL and GLO REP
12/2020	**ACRI** ▪ GLO REP: PFT provided for the period [2002-present] ▪ GLO NRT/REP: update of chlorophyll, ZSD, PP, KD products (including uncertainties for L3 ZSD) ▪ ATL NRT/REP: update of chlorophyll products ▪ GLO and ATL: extension of 6 months of the REP datasets **CNR** ▪ MED and BS NRT/DT/REP – inclusion of NOAA20-VIIRS in multi-sensor products ▪ MED NRT/DT/REP – update of chlorophyll algorithm

	▪ MED and BS REP – new reprocessed products based on NRT/DT multi-sensor products ▪ MED and BS REP – 3 new IOP datasets introduced: bbp443, aph443 and adg443 **PML** ▪ ATL and ARC NRT/DT/REP – Single-sensor MODIS ADG, APH and ATOT and VIIRS SPM products have been withdrawn from the catalogue, they have been superseded by multi-sensor products based on OC-CCI ▪ ATL and ARC NRT/DT/REP – update of chlorophyll oc5cci algorithm ▪ GLO REP – new reprocessed products based on ESA CCI v5.0
5/2021	**ACRI** ▪ GLO NRT/REP: PFT provided with an uncertainty field ▪ GLO NRT: a 300 meters spatial resolution coastal chlorophyll product based on OLCI-S3A/S3B merging is now disseminated **CNR** ▪ MED and BS REP – daily multi-sensor L4 interpolated (cloud free) product is now available as a REP [1997-2020] ▪ MED, BS and BAL NRT – 300 meters spatial resolution OLCI-S3A/S3B merging products are now disseminated ▪ BS and BAL REP – introducing PFT dataset in CHL product ▪ BAL REP – update of chlorophyll algorithm and rrs datasets **PML** ▪ ARC REP: PFT provided ▪ ARC and ATL REP: multi-sensor SPM provided ▪ ARC and ATL NRT: 300 meters spatial resolution OLCI-S3A/S3B merging products are now disseminated
12/2021	**ACRI** ▪ The OLCI-S3A single sensor datasets are replaced by an OLCI S3A & S3B merged product (NRT and REP) based on a new EUMETSAT upstream reprocessing. **CNR** ▪ New product for L4 NRT BAL CHL **PML** ▪ ARC, ATL and GLO REP: PFT uncertainties provided.
07/2022	**ACRI, CNR, PML** ▪ PUM updated to reflect catalogue reorganization and new template
12/2022	**ACRI**

	▪ GLOB and ATL are updated to use the EUMETSAT collection 3 upstream reprocessing . ▪ GlO 300 m products are now available from 2016 with additional monthly and reflectances datasets. **PML** ▪ ARC, ATL and GLO NRT and MY, OLCI-S3B now included in the multi products ▪ ARC and ATL, new 300 metre spatial resolution OLCI-S3A/S3B MY products now disseminated **CNR** ▪ MED, BLK and BAL NRT – OLCI-S3B included in the multi products ▪ MED, BLK and BAL NRT – Introducing PFT in PLANKTON dataset ▪ MED, BLK and BAL MY – 300 meters spatial resolution OLCI-S3A&S3B merging products (collection 3.01) are now available from 2016 with additional monthly and reflectances datasets ▪ BAL NRT – Updated with new atmospheric correction and new chlorophyll algorithm ▪ BAL MY – Updated with new ESA OC-CCIv5 data
03/2023	**ACRI** ▪ Products OCEANCOLOUR_ATL_BGC_L3_NRT_009_111 and OCEANCOLOUR_ATL_BGC_L3_MY_009_113 formerly produced by PML PU have been reintroduced to the catalogue by ACRI PU. They are now produced with the GlobColour processing chain. **CNR** ▪ Uptake of all ARC PML products excluding multi sensor datasets **BC** ▪ Uptake of former PML global products OCEANCOLOUR_GLO_BGC_L3_MY_009_107 and OCEANCOLOUR_GLO_BGC_L4_MY_009_108 with no change, making use of CCI v6 (including NASA R2018.0)
11/2023	**ACRI** • MY products reprocessed based on NASA R2022 re-processing, MERIS 4th re-processing, and OLCI collection 3.01. • Update of the BBP, CDM, and PP algorithms. • Addition of PFT and PP datasets in the Atlantic region.

	CNR ▪ MED MY: Introduction of Primary Production dataset ▪ MED and BLK MY: new multi-sensor reprocessed products based on : NASA R2022 re-processing, MERIS 4th re-processing, and OLCI collection 3.01
11/2024	**CNR** ▪ MED NRT: Introduction of Primary Production dataset and update of PFT dataset ▪ MED MY: update of PFT dataset ▪ BLK NRT/MY: Introduction of Primary Production dataset ▪ BAL MY: Introduction of Primary Production dataset
03/2026	**ACRI** • Introduction of VIIRS-J2 (NOAA21) and retirement of MODIS-Aqua.

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Information

II.1.1 Product Levels

There are two product levels that are disseminated by Copernicus Marine Service OCTAC: L3 and L4. Input data (Level-2 or Level-1) are the products downloaded by OCTAC PUs from upstream data providers.

Input data

- Level 1 (L1) are raw data formatted at full instrument resolution, time-referenced, and annotated with ancillary information, including radiometric and geometric calibration coefficients and georeferencing parameters (e.g., platform ephemeris) computed and appended but not applied to Level 0 data
- Level 2 data consist of derived geophysical variables at the same resolution as the source Level 1 data acquired by the ocean colour remote sensors.

L2 are the input data of all products derived by ACRI and CNR PUs, and most of the products from the PML PU. L1 data are primarily used in OC-CCI related processing, which allows greater control and choice of algorithms. It may also be used in limited circumstances, such as working around any lengthy upstream L2 outage.

L3 Products

- L3 products are the daily composite and can be of any spatial resolution, typically 300 meters, 1km and 4 km

L4 Products

- Level 4 data result from analyses of L3 data (e.g., variables derived from multiple measurements).
- L4 are those products for which a temporal averaging method or an interpolation procedure is applied to fill in missing data values. Temporal averaging is performed on a monthly basis. The L4 daily products is also called “interpolated” or “cloud free” products.

II.1.2 Product processing mode and temporal coverage

The OCTAC processes the L3 and L4 ocean colour products in two different processing modes:

- Operational mode
- Offline processing mode

These two processing modes are associated to the Copernicus Marine Service catalogue products definition:

- The Near real Time (NRT) products
- The multi-year time series of reprocessed (MY) products

NRT and MY products could be generated using different input data. As a consequence, the NRT and MY products could be characterized by different scientific quality. A more detailed description of these product versions is provided below.

II.1.3 Operational Near real Time (NRT) products

The **NRT products** (defined "nrt", in the product file name) are operationally produced every day and provide the best estimate of the ocean colour variables at the time of processing. These products are generated using the best auxiliary data available (meteorological data) at the time of the processing. NRT data are meant for providing the user with products as soon as possible.

Due to unavailability of ancillary data (precision orbital data and meteorological fields used for atmospheric correction), used to produce L2 data, NRT products could have lower scientific quality. Nevertheless, these NRT data are very useful for NRT coastal application, water quality monitoring, fishery application, and in situ data acquisition. This version of the product is delivered within one day from the satellite acquisition, compatible with the less-than-24 hours requirement.

II.1.4 Reprocessed (MY) products

The **MY products** (defined "my", in the product file name) are re-processed once for a long period (some years) using a consolidated and consistent input dataset with a unique processing software configuration. So, MY products should be used for climatic study or for analysis of the inter-annual variability of the ocean parameter under study.

- MY products provide a multi-year time series of the OC observations in the past time, the entire time series is available on line accessing the OCTAC DU from the Copernicus serving desk
- MY products are available at both regional and global scales.
- The Copernicus-GlobColour and OC-CCI are the main initiatives in the world providing a long term series of level 3 Observation Ocean Colour products at global level based on a multi sensor approach with a 4km resolution. While both datasets aim to provide high-quality merged datasets, Copernicus-GlobColour supports continuity with the NRT production (and thus the unavoidable risk of inconsistencies in the time series due to recalibrations or processing changes in its upstream data, until a full reprocessing is done on a yearly basis to restore consistency). On the other hand, OC-CCI targets "climate quality" consistency with minimal inter-sensor bias, with the unavoidable cost of being unable to provide this consistency up to the current date (e.g. with NRT). In V4, the CCI algorithms are applied to delayed-time NRT data for the regions, reducing the difference to around a month. These two datasets therefore provide different and complementary characteristics that are useful to different users – a service offering an NRT stream may

opt for the GC MY in order to be able to identify anomalies in yesterday's NRT, while a study investigating delayed-time NRT, longer subtle changes over time or deriving a historical measure for later use may opt for OC-CCI or Copernicus-GlobColour. The differences between the two initiatives are detailed in the Section V

- The regional MY products are produced to improve the Chlorophyll retrieval accuracy and to provide a finer resolution (1Km) to address regional user needs. To this aim, for ARC and ATL, the OC-CCI reflectances at 1km are ingested and then the same regional bio-optical algorithms used in NRT production are applied. For the OC-CCI reflectances at 1km are ingested and then a dedicated regional bio-optical algorithm is applied, while for BLK and MED the same processing chain is applied for the generation of the multi-sensor reflectances at 1km and same regional bio-optical algorithms.
- The users will be advised by Copernicus Marine Service each time a new version of the MY products will be available

MY time series are extended, with the **MY interim** (MYINT) production (defined "myint", in the product file name) using the same configuration of the previous full reprocessing, to fill the gap between MY and NRT products.

- the MYINT are produced within 31 days from sensing using consolidated L2 products to obtain consolidated products as soon as ancillary data are available. As EUMETSAT and NASA currently have no matching schedules for L2 consolidation (i.e. EUMETSAT about 8 days while NASA about 2 months), MYINT production will occur twice for the multi-sensor datasets and only once for the OLCI datasets. The errors associated to the MYINT files are in general lower than the NRT one. The file name will reflect this change to ensure that the users are unequivocally aware that they are actually downloading OC interim products
- the files produced with MYINT production will get replaced with MY files only when a full reprocessing is carried out as part of an EiS

II.2 Details of the products

The general information of the OC global and regional products is presented in the following subsections. Table 2 summarizes the OCTAC catalogue codes for each product and region.

CMEMS Region	Multi-Sensor 1km (Regions), 4km (GLO+ARC)				Sentinel-3 OLCI A+B 300m (Regions), 4km & 300m (GLO)			
	NRT		MY		NRT		MY	
	L3	L4	L3	L4	L3	L4	L3	L4
GLO (ACRI)	101	102	103	104	101	102	103	104
GLO (BC)			107	108				
ATL	111	116	113	118	111		113	

ARC			123	124	121	122	123	124
BAL	-	-	133	134	131	132	133	134
MED	141	142	143	144	141	142	143	144
BLK	151	152	153	154	151	152	153	154

Table 2 : Real-Time and Multi-Years product numbers: *OCEANCOLOUR_<region>_BGC_<L3/L4>_<NRT/MY>_009_1XX*

Detailed information on the systems and products are available on Copernicus Mrine web site: https://marine.copernicus.eu/.

II.2.1 Global Ocean (Copernicus-GlobColour)

These products are provided by ACRI-ST Company (Sophia Antipolis, France) and are based on the Copernicus-GlobColour project (see section V). Products distributed in the frame of CMEMS are a subset of the larger Copernicus-GlobColour database (more than 40 variables available at 100, 25, 4, 1km and 300 meters.

Details of Copernicus Marine products provided by ACRI-ST are provided below. Upstream sensors used are SeaWiFS, MODIS-Aqua & Terra, MERIS, VIIRS-SNPP & JPSS1, OLCI-S3A & S3B.

Product Lines	OCEANCOLOUR_GLO_BGC_L3_NRT_009_101 OCEANCOLOUR_GLO_BGC_L4_NRT_009_102 OCEANCOLOUR_GLO_BGC_L3_MY_009_103 OCEANCOLOUR_GLO_BGC_L4_MY_009_104
Geographical coverage	180°W - 180°E; 90°S - 90°N
Variables (for multi sensors products based on SeaWIFS, MERIS, MODIS-A, MODIS-T, VIIRS-SNPP & JPPS1, OLCI-S3A & S3B)	▪ Chlorophyll-a concentration (CHL) ▪ Remote Sensing Reflectance at 412, 443, 490, 555, 670 nm (RRS) ▪ Diffuse attenuation coefficient at 490 nm (KD490) ▪ Particulate backscattering coef. at 443 nm (BBP) ▪ Absorption coef. due to cdom and non-pigmented particles at 443 nm (CDM) ▪ Secchi disk depth Transparency (ZSD) ▪ Non-Algal Suspended Particle Matter (SPM) ▪ Phytoplankton Functional Type (PFT) ▪ Primary Production (PP)
Variables (for multi sensors products based on Sentinel3A+3B OLCI at high resolution)	• Chlorophyll-a concentration (CHL) • Remote Sensing Reflectance at 400, 412, 443, 490, 510, 560, 620, 665, 674, 681, 709 nm (RRS) • Diffuse attenuation coefficient at 490 nm (KD490)

	• Norm of the horizontal gradient of the Chlorophyll-a concentration (CHL_gradient)
Product Type	Observations
Available time series	Multi sensor NRT/MYINT: [start from the earliest date not covered by MY products - present] Multi sensor MY: start from September 1997 (multi) or April 2016 (olci) until about six months before the date of the full reprocessing (see product improvements pages https://marine.copernicus.eu/user-corner/user-notification-service?field_category=27). PFT are not available before 2002 (when only SEAWIFS sensor available) It should be noted that some daily products are missing due to the non-availability of upstream (mainly SeaWIFS input data).
Temporal resolution	Daily, monthly and daily-climatology
Target delivery time	NRT – daily: 1 day after satellite acquisition at 22:00 UTC MYINT – daily: 31 days after satellite acquisition at 22:00 UTC NRT – Monthly: 1 day after end of the period at 22:00 UTC MYINT - Monthly: 31 days after end of the period at 22:00 UTC
Delivery mechanism	Marine Data Store
Horizontal resolution	• 4km(*) for Multi-sensor • 4km(*) for Sentinel3-OLCI (S3A & S3B merged) • 300m(*) for the global coastal product (up to 200 km from the coastline, extended to 20W in the European Atlantic.)
Number of vertical levels	From 0.0m to 0.0m (CRS=EPSG: 5714)
Format	NetCDF CF1.4

Table 3 : GLO-OCTAC (Copernicus-GlobColour) Real-Time and Multi-Years products

(*) the products are distributed under "plate carrée" geometrical system. This means that it relies on a regular angular resolution in latitude and longitude. This, naturally, due to Earth's shape, are translated in spatial resolution with variable projected distance resolution. The "4km"

and "300 m" indication above about « horizontal » are therefore indicative (as they cannot be ensured at a global level with uniform angular step).

The "4 km" label means that the pixel size for the Global product at the equator is about 4x4km leading to a 4 km stable vertical resolution for all longitude and a decreasing horizontal resolution when latitude is in/decreasing towards poles.

The "300 m" label means that the pixel size for the Global products (which is anyway built from 300 m OLCI product) is restituted at about 600x600 m resolution at equator leading to a 600 m stable resolution for all longitude and a decreasing horizontal resolution when latitude is in/decreasing towards poles. The 300m resolution is effective around 47N/S. This is done to ensure, from initial 300m OLCI products, the best compromise between noise filtering and the best spatial resolution.

II.2.2 Global Ocean (OC-CCI)

The Ocean Colour Climate Change Initiative is part of an ESA programme to produce "climate-grade" Essential Climate Variables based on satellite data; in OC-CCI v6's case, a merger of SeaWiFS, VIIRS, MERIS, MODIS-Aqua, OLCI-S3A and OLCI-S3B using the best-performing atmospheric correction and chlorophyll algorithms, with a temporally-weighted bias correction aiming to minimise differences between sensors. There is an annual update of algorithms and a bi-annual update of the time series, which was incorporated by PML into the CMEMS distribution as GLO MY products and now overtaken by BC. Further information and optional acknowledgement text can be found on the OC-CCI website (https://www.oceancolour.org/).

Product Lines	OCEANCOLOUR_GLO_BGC_L3_MY_009_107 OCEANCOLOUR_GLO_BGC_L4_MY_009_108
Geographical coverage	180°W – 180°E; 90°S – 90°N
Variables (for multi sensors products based on SeaWIFS, MERIS, MODIS-Aqua, NPP-VIIRS, Sentinel3A-OLCI, Sentinel3B-OLCI)	• Chlorophyll-a concentration (L3 daily and L4 monthly) • Phytoplankton Functional Type (PFT) ○ expressed as Chlorophyll a concentration in sea water (mg m-3), includes the following variables: MICRO (Micro-phytoplankton), NANO (Nano-phytoplankton) and PICO (Pico-phytoplankton) also known as "Phytoplankton Size Classes" (PSCs) • Remote Sensing Reflectance at 412, 443, 490, 510, 560, 665 nm
Product Type	Observations
Available time series	Start from September 1997 until about six months before the date of the full reprocessing

	It should be noted that some daily products are missing during the period [1997-2002] due to the non-availability of the SeaWIFS input data.
Temporal resolution	Daily, monthly
Target delivery time	updated as needed
Delivery mechanism	Marine Data Store
Horizontal resolution	▪ 4 Km
Number of vertical levels	From 0.0m to 0.0m (CRS=EPSG: 5714)
Format	NetCDF CF1.7

Table 4 : GLO-OCTAC (OC-CCI) Multi-Years products

II.2.3 Atlantic

Details about products provided by ACRI-ST (daily and monthly) for the ATL region are shown below.

Product Lines	OCEANCOLOUR_ATL_BGC_L3_NRT_009_111 OCEANCOLOUR_ATL_BGC_L3_MY_009_113
Geographical coverage	46°W – 13°E; 20°N – 66°N. Data is available 200km from the coastline, with an extension to 20W from mainland Europe..
Variables (for multi sensors products based on SeaWIFS, MERIS, MODIS-A, MODIS-T, VIIRS-SNPP & JPPS1, OLCI-S3A & S3B)	▪ Chlorophyll-a concentration (CHL) ▪ Remote Sensing Reflectance at 412, 443, 490, 555, 670 nm (RRS) ▪ Diffuse attenuation coefficient at 490 nm (KD490) ▪ Particulate backscattering coef. at 443 nm (BBP) ▪ Absorption coef. due to cdom and non-pigmented particles at 443 nm (CDM) ▪ Secchi disk depth Transparency (ZSD) ▪ Non-Algal Suspended Particle Matter (SPM) ▪ Phytoplankton Functional Type (PFT)
Variables (for multi sensors products based on high resolution Sentinel3A-OLCI and Sentinel3B-OLCI)	● Chlorophyll-a concentration (CHL) ● Remote Sensing Reflectance at 400, 412, 443, 490, 510, 560, 620, 665, 674, 681, 709 nm (RRS) ● Diffuse attenuation coefficient at 490 nm (KD490)

	• Norm of the horizontal gradient of the Chlorophyll-a concentration (CHL_gradient)
Product Type	Observations
Available time series	Multi sensor NRT/MYINT: [start from the earliest date not covered by MY products – present] Multi sensor MY: start from September 1997 (multi) or April 2016 (olci) until about six months before the date of the full reprocessing (see product improvements pages https://marine.copernicus.eu/user-corner/user-notification-service?field_category=27). PFT are not available before 2002 (when only SEAWIFS sensor available) It should be noted that some daily products are missing due to the non-availability of upstream (mainly SeaWIFS input data).
Temporal resolution	Daily
Target delivery time	NRT – daily: 1 day after satellite acquisition at 22:00 UTC NRT - Monthly: 3 days after end of the period at 22:00 UTC MYINT – daily: 31 days after satellite acquisition at 22:00 UTC MYINT – Monthly: 31 days after end of the period at 22:00 UTC
Delivery mechanism	Marine Data Store
Horizontal resolution	▪ 1 km for Multi-sensor ▪ 300 m for Sentinel3-OLCI
Number of vertical levels	From 0.0m to 0.0m (CRS=EPSG: 5714)
Format	NetCDF CF1.7

Table 5 : ATL-OCTAC Real-Time and Multi-Years products

The multi sensor MY product provides Rrs data compatible with the OC-CCI 4km GLO MY, but reprocessed from scratch at 1km resolution for the regional areas. The CHL product provides the regional algorithm (OC5CCI) created using these Rrs. This is fundamentally identical to the NRT multi-sensor product in terms of the algorithm used, but differs in that this product comes from the time period covered by OC-CCI and thus has a somewhat higher level of quality control.

The Atlantic L4 interpolated Chlorophyll product (cloud-free) is provided by ACRI-ST Company (Sophia Antipolis, France) and is based on the Copernicus-GlobColour processor (see section V). Details of this product are provided below.

Product Lines	OCEANCOLOUR_ATL_BGC_L4_NRT_009_116 OCEANCOLOUR_ATL_BGC_L4_MY_009_118
Geographical coverage	46°W – 13°E; 20°N – 66°N
Variables (based on SeaWIFS, MERIS, MODIS-A, MODIS-T, VIIRS-SNPP & JPPS1, OLCI-S3A & S3B)	• Chlorophyll-a concentration (CHL) • Phytoplankton Functional Type (PFT) • Primary Production (PP)
Product Type	Observations
Available time series	Multi sensor NRT/MYINT: [start from the earliest date not covered by MY products - present] Multi sensor MY: start from September 1997 (multi) or April 2016 (olci) until about six months before the date of the full reprocessing (see product improvements pages https://marine.copernicus.eu/user-corner/user-notification-service?field_category=27). PFT are not available before 2002 (when only SEAWIFS sensor available) It should be noted that some daily products are missing due to the non-availability of upstream (mainly SeaWIFS input data).
Temporal resolution	Daily, Monthly
Target delivery time	NRT – daily: 1 day after satellite acquisition at 22:00 UTC MYINT – daily: 31 days after satellite acquisition at 22:00 UTC (except Interpolated products at 36 days after satellite acquisition at 22:00 UTC) MY: updated as needed
Delivery mechanism	Marine Data Store
Horizontal resolution	• 1 Km
Number of vertical levels	From 0.0m to 0.0m (CRS=EPSG: 5714)
Format	NetCDF CF1.4

Table 6 : ATL-OCTAC Real-Time and Multi-Years products (interpolated)

II.2.4 Artic Region

Details about products provided by CNR (daily and monthly) for the Arctic region are shown below.

Product Lines	OCEANCOLOUR_ARC_BGC_L3_NRT_009_121 OCEANCOLOUR_ARC_BGC_L4_NRT_009_122 OCEANCOLOUR_ARC_BGC_L3_MY_009_123 OCEANCOLOUR_ARC_BGC_L4_MY_009_124
Geographical coverage	180°W – 180°E; 66°N – 90°N
Variables (for multi sensors products based on high resolution Sentinel3A-OLCI and Sentinel3B-OLCI)	• Chlorophyll-a concentration (L3 daily and L4 monthly) • Remote Sensing Reflectance at 400, 412, 443, 490, 510, 560, 620, 665, 674, 681, 709 nm • Diffuse attenuation coefficient at 490 nm • Absorption coeff. Due to gelbstoff and detrital material at 443 nm • Absorption coeff. Due to phytoplankton at 443 nm • Particulate back-scattering coeff. At 443 nm
Product Type	Observations
Available time series	OLCI-Sentinel3 NRT/MYINT: [start from the earliest date not covered by MY products - present] OLCI-Sentinel3 MY: start from 25 April 2016 until about six months before the date of the full reprocessing Multi sensor MY: start from September 1997 until about six months before the date of the full reprocessing It should be noted that some Multi sensor MY daily products are missing during the period [1997-2002] due to the non-availability of the SeaWIFS input data.
Temporal resolution	Daily, monthly
Target delivery time	NRT – daily: 1 day after satellite acquisition at 18:00 UTC NRT - Monthly: 3 days after end of the period at 18:00 UTC MYINT – daily: 31 days after satellite acquisition at 18:00 UTC MYINT – Monthly: 31 days after end of the period at 20:00 UTC
Delivery mechanism	Marine Data Store

Horizontal resolution	▪ 4km for Multi-sensor ▪ 300 m for Sentinel3-OLCI
Number of vertical levels	From 0.0m to 0.0m (CRS=EPSG: 5714)
Format	NetCDF CF1.7

Table 7 : ARC-OCTAC Real-Time and Multi-Years products

II.2.5 Baltic Sea

Details about products provided by CNR with daily temporal resolution for the Baltic Sea are shown below.

Product Lines	OCEANCOLOUR_BAL_BGC_L3_NRT_009_131 OCEANCOLOUR_BAL_BGC_L4_NRT_009_132 OCEANCOLOUR_BAL_BGC_L3_MY_009_133 OCEANCOLOUR_BAL_BGC_L4_MY_009_134
Geographical coverage	9.25°E – 30.25°E; 53.25°N – 65.85°N
Variables (for multi sensors products based on OLCI-Sentinel3A&B at high resolution)	● Chlorophyll-a concentration ([RD11],[RD12],[RD13],[RD14]) ● Phytoplankton Functional Type (PFT) ○ expressed as Chlorophyll a concentration in sea water (mg m-3), includes the following variables: DIATO (Diatoms), DINO (Dinophytes or Dinoflagellates), CRYPTO (Cryptophytes), GREEN (Green algae & Prochlorophytes) and PROKAR (Prokaryotes); MICRO (Micro-phytoplankton), NANO (Nano-phytoplankton) and PICO (Pico-phytoplankton) also known as "Phytoplankton Size Classes" (PSCs). MICRO consist of DIATO and DINO, NANO of CRYPTO and half of the GREEN group and PICO includes half GREEN and PROKAR. Note: the development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation ● Remote Sensing Reflectance at 400, 412, 443, 490, 510, 560, 620, 665, 674, 681, 709 nm ● Diffuse attenuation coefficient at 490 nm ● Absorption coeff. Due to gelbstoff and detrital material at 443 nm

	• Absorption coeff. Due to phytoplankton at 443 nm • Particulate back-scattering coeff. at 443 nm • Integrated Primary Production
Variables (for multi sensors products based on SeaWIFS, MERIS, MODIS-Aqua, NPP-VIIRS and OLCI-Sentinel3A)	• Chlorophyll-a concentration ([RD11],[RD12],[RD13],[RD14]) • Phytoplankton Functional Type (PFT) ○ expressed as Chlorophyll a concentration in sea water (mg m-3), includes the following variables: DIATO (Diatoms), DINO (Dinophytes or Dinoflagellates), CRYPTO (Cryptophytes), GREEN (Green algae & Prochlorophytes) and PROKAR (Prokaryotes); MICRO (Micro-phytoplankton), NANO (Nano-phytoplankton) and PICO (Pico-phytoplankton) also known as "Phytoplankton Size Classes" (PSCs). MICRO consist of DIATO and DINO, NANO of CRYPTO and half of the GREEN group and PICO includes half GREEN and PROKAR. Note: the development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation • Remote Sensing Reflectance at 412, 443, 490, 555, 670 nm • Diffuse attenuation coefficient at 490 nm
Product Type	Observations
Available time series	OLCI-Sentinel3 NRT/MYINT: [start from the earliest date not covered by MY products - present] OLCI-Sentinel3 MY: start from 25 April 2016 until about six months before the date of the full reprocessing Multi sensor MY: start from September 1997 until about six months before the date of the full reprocessing It should be noted that some Multi sensor MY daily products are missing during the period [1997-2002] due to the non-availability of the SeaWIFS input data.
Temporal resolution	Daily, monthly
Target delivery time	NRT – Daily: 1 day after satellite acquisition at 20:00 UTC MYINT – Daily: 31 days after satellite acquisition at 18:00 UTC

	NRT – Monthly: 1 days after end of the period at 20:00 UTC MYINT – Monthly: 31 days after end of the period at 18:00 UTC
Delivery mechanism	Marine Data Store
Horizontal resolution	▪ 1km for Multi-sensor ▪ 300 m for OLCI-Sentinel3
Number of vertical levels	From 0.0m to 0.0m (CRS=EPSG: 5714)
Format	NetCDF CF1.4

Table 8 : BAL-OCTAC Real-Time and Multi-Year products

II.2.6 Mediterranean Sea

Details about products provided by CNR (daily, monthly and interpolated) for the MED region are shown below.

Product Lines	OCEANCOLOUR_MED_BGC_L3_NRT_009_141 OCEANCOLOUR_MED_BGC_L4_NRT_009_142 OCEANCOLOUR_MED_BGC_L3_MY_009_143 OCEANCOLOUR_MED_BGC_L4_MY_009_144
Geographical coverage	6°W – 36.5°E; 30.0°N – 46.0°N
Variables (for multi sensors products based on SeaWIFS, MERIS, MODIS-Aqua, NPP-VIIRS, NOAA20-VIIRS and OLCI-Sentinel3A&B)	● Chlorophyll-a concentration ○ L3 daily and L4 monthly ([RD1],[RD2],[RD3],[RD6]) ○ L4 interpolated (cloud-free) ([RD1],[RD2],[RD4],[RD6][RD10]) ● Phytoplankton Functional Type (PFT) ○ expressed as Chlorophyll a concentration in sea water (mg m-3), includes the following variables: DIATO (Diatoms), DINO (Dinophytes or Dinoflagellates), CRYPTO (Cryptophytes), HAPTO (Haptophytes or Coccolitophores), GREEN (Green algae & Prochlorophytes) and PROKAR (Prokaryotes); MICRO (Micro-phytoplankton), NANO (Nano-phytoplankton) and PICO (Pico-phytoplankton) also known as "Phytoplankton Size Classes" (PSCs). MICRO consist of DIATO and DINO, NANO of CRYPTO and HAPTO and PICO includes GREEN and PROKAR. In the ultra-oligotrophic condition HAPTO were divided between NANO and PICO following the adjustment of [RD15]. Note: the development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT

	estimate provided in other products (GLO and ATL). For more details see the relevant documentation. ([RD7]) • Remote Sensing Reflectance at 412, 443, 490, 555, 670 nm • Diffuse attenuation coefficient at 490 nm • Absorption coeff. Due to gelbstoff and detrital material at 443 nm • Absorption coeff. Due to phytoplankton at 443 nm • Particulate back-scattering coeff. At 443 nm • Integrated Primary Production
Variables (for multi sensors products based on OLCI-Sentinel3A&B at high resolution)	• Chlorophyll-a concentration ([RD1],[RD2],[RD3],[RD6]) • Remote Sensing Reflectance at 400, 412, 443, 490, 510, 560, 620, 665, 674, 681, 709 nm • Diffuse attenuation coefficient at 490 nm
Product Type	Observations
Available time series	OLCI-Sentinel3 NRT/MYINT: [start from the earliest date not covered by MY products – present] Multi sensor NRT/MYINT: [start from the earliest date not covered by MY products – present] OLCI-Sentinel3 MY: start from 25 April 2016 until about six months before the date of the full reprocessing Multi sensor MY: start from September 1997 until about six months before the date of the full reprocessing It should be noted that some Multi sensor MY daily products are missing during the period [1997-2002] due to the non-availability of the SeaWIFS input data.
Temporal resolution	Daily, monthly and daily-climatology
Target delivery time	NRT – Daily: 1 day after satellite acquisition at 20:00 UTC MYINT – Daily: 31 days after satellite acquisition at 18:00 UTC (except Interpolated products at 36 days after satellite acquisition at 18:00 UTC) NRT – Monthly: 1 days after end of the period at 20:00 UTC MYINT – Monthly: 31 days after end of the period at 18:00 UTC
Delivery mechanism	Marine Data Store
Horizontal resolution	▪ 1 km for Multi-sensor (4 km for Primary Production)

	▪ 300 m for OLCI-Sentinel3
Number of vertical levels	From 0.0m to 0.0m (CRS=EPSG: 5714)
Format	NetCDF CF1.4

Table 9 : MED-OCTAC Real-Time and Multi-Year products

Satellite climatology is obtained from twenty-five years of satellite, multi-sensor, observations, using an update version of the MedOC4.2018 regional algorithm for Chl ([RD6]) (MedOC4.2020), with a nominal spatial resolution of 1 km. These climatology maps have been created using the data falling into a moving temporal window of ± 5 days. One of the main purposes of a climatology field is to serve as reference, and as such it is expected to be as reliable as possible, thus avoiding biases caused by single incorrect pixel values. To overcome these possible biases, a filtering procedure has been applied to the entire SeaWiFS time series, by removing all isolated pixels and by filling in all isolated missing pixels using the near-neighbourhood approach. The resulting daily climatology time series includes the average, the median and the standard deviation (STD) on a pixel-by-pixel basis.

II.2.7 Black Sea

Details about products provided by CNR (daily, monthly and interpolated) for the BLK region are shown below.

Product Lines	OCEANCOLOUR_BLK_BGC_L3_NRT_009_151 OCEANCOLOUR_BLK_BGC_L4_NRT_009_152 OCEANCOLOUR_BLK_BGC_L3_MY_009_153 OCEANCOLOUR_BLK_BGC_L4_MY_009_154
Geographical coverage	26.5°E – 42.0°E; 40.0°N – 48.0°N
Variables (for multi sensors products based on SeaWIFS, MERIS, MODIS-Aqua, NPP-VIIRS, NOAA20-VIIRS and OLCI-Sentinel3A&B)	● Chlorophyll-a concentration ○ L3 daily and L4 monthly ([RD6],[RD8],[RD9]) ○ L4 interpolated (cloud-free) ([RD1],[RD4],[RD6],[RD8],[RD9],[RD10]) ● Phytoplankton Functional Type (PFT) ○ expressed as Chlorophyll a concentration in sea water (mg m-3), includes the following variables: MICRO (Micro-phytoplankton), NANO (Nano-phytoplankton) and PICO (Pico-phytoplankton) also known as “Phytoplankton Size Classes” (PSCs). MICRO consist of Diatoms (DIATO) and Dinophytes or Dinoflagellates (DINO), NANO of Cryptophytes (CRYPTO), Haptophytes (HAPTO) and half

	of "Green algae & Prochlorophytes" group (GREEN) and PICO includes half GREEN and Prokaryotes (PROKAR). Note: the development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation ● Remote Sensing Reflectance at 412, 443, 490, 555, 670 nm ● Diffuse attenuation coefficient at 490 nm ● Absorption coeff. Due to gelbstoff and detrital material at 443 nm ● Absorption coeff. Due to phytoplankton at 443 nm ● Particulate back-scattering coeff. At 443 nm ● Integrated Primary Production
Variables (for multi sensors products based on OLCI-Sentinel3A&B at high resolution)	● Chlorophyll-a concentration ([RD6],[RD8],[RD9]) ● Remote Sensing Reflectance at 400, 412, 443, 490, 510, 560, 620, 665, 674, 681, 709 nm ● Diffuse attenuation coefficient at 490 nm
Product Type	Observations
Available time series	OLCI-Sentinel3 NRT/MYINT: [start from the earliest date not covered by MY products – present] Multi sensor NRT/MYINT: [start from the earliest date not covered by MY products – present] OLCI-Sentinel3 MY: start from 25 April 2016 until about six months before the date of the full reprocessing Multi sensor MY: start from September 1997 until about six months before the date of the full reprocessing It should be noted that some Multi sensor MY daily products are missing during the period [1997-2002] due to the non-availability of the SeaWIFS input data.
Temporal resolution	Daily, monthly and daily-climatology
Target delivery time	NRT – Daily: 1 day after satellite acquisition at 20:00 UTC MYINT – Daily: 31 days after satellite acquisition at 18:00 UTC (except Interpolated products at 36 days after satellite acquisition at 18:00 UTC)

	NRT – Monthly: 1 days after end of the period at 20:00 UTC MYINT – Monthly: 31 days after end of the period at 18:00 UTC
Delivery mechanism	Marine Data Store
Horizontal resolution	▪ 1 km for Multi-sensor ▪ 300 m for OLCI-Sentinel3
Number of vertical levels	From 0.0m to 0.0m (CRS=EPSG: 5714)
Format	NetCDF CF1.4

Table 10 : BLK-OCTAC Real-Time and Multi-Years products

Satellite climatology is obtained from twenty-five years of satellite, multi-sensor, observations, using the BSAlg regional algorithm for Chl ([RD8],[RD9]), with a nominal spatial resolution of 1 km. These climatology maps have been created using the data falling into a moving temporal window of ± 5 days. One of the main purposes of a climatology field is to serve as reference, and as such it is expected to be as reliable as possible, thus avoiding biases caused by single incorrect pixel values. To overcome these possible biases, a filtering procedure has been applied to the entire SeaWiFS time series, by removing all isolated pixels and by filling in all isolated missing pixels using the near-neighbourhood approach. The resulting daily climatology time series includes the average, the median and the standard deviation (STD) on a pixel-by-pixel basis.

II.3 Details of datasets

The details of the OC global and regional datasets are presented in the following subsections. Each subsection refers to different ocean region. It is important to underline that the Ocean Colour datasets are classified also taking into account the satellite product level definition (L3 or L4) and the product temporal coverage.

II.3.1 Global Ocean (Copernicus-GlobColour)

II.3.1.1 OCEANCOLOUR_GLO_BGC_L3_NRT_009_101 (Copernicus-GlobColour)

OCEANCOLOUR_GLO_BGC_L3_NRT_009_101	
Dataset: **cmems_obs-oc_glo_bgc-reflectance_nrt_l3-olci-4km_P1D** **cmems_obs-oc_glo_bgc-reflectance_nrt_l3-olci-300m_P1D**	
Variables name in the NetCDF file	RRS400

Long_name	Fully normalised remote sensing reflectance at 400 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS412
Long_name	Fully normalised remote sensing reflectance at 412 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Fully normalised remote sensing reflectance at 443 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Fully normalised remote sensing reflectance at 490 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Fully normalised remote sensing reflectance at 510 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS560
Long_name	Fully normalised remote sensing reflectance at 560 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS620
Long_name	Fully normalised remote sensing reflectance at 620 nm – Mean of the binned pixels

Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Fully normalised remote sensing reflectance at 665 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS674
Long_name	Fully normalised remote sensing reflectance at 674 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS681
Long_name	Fully normalised remote sensing reflectance at 681 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS709
Long_name	Fully normalised remote sensing reflectance at 709 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-reflectance_nrt_l3-multi-4km_P1D	
Variables name in the NetCDF file	RRS412
Long_name	Fully normalised remote sensing reflectance at 412 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Fully normalised remote sensing reflectance at 443 nm – Mean of the binned pixels

Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Fully normalised remote sensing reflectance at 490 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS555
Long_name	Fully normalised remote sensing reflectance at 555 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS670
Long_name	Fully normalised remote sensing reflectance at 670 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-plankton_nrt_l3-olci-300m_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-plankton_nrt_l3-olci-4km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	CHL_gradient
Long_name	Norm of the horizontal gradient of the chlorophyll concentration
Standard_name:	magnitude_of_horizontal_gradient_of_mass_concentration_of_chlorophyll_a_in_sea_water

Units	milligram m-3 km-1
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-plankton_nrt_l3-multi-4km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll-a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	DIATO
Long_name	Diatoms Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_diatoms_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	DINO
Long_name	Dinophytes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_dinophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	GREEN
Long_name	Green Algae Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_greenalgae_and_prochlorophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	HAPTO
Long_name	Haptophytes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_haptophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PROCHLO
Long_name	Prochlorococcus Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_prochlorococcus_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PROKAR
Long_name	Prokaryotes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_prokaryotes_expressed_as_chlorophyll_in_sea_water

Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	MICRO
Long_name	Microphytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_microphytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	NANO
Long_name	Nanophytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_nanophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PICO
Long_name	Picophytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_picophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-transp_nrt_l3-olci-4km_P1D	
Variables name in the NetCDF file	KD490
Long_name	Diffuse attenuation coefficient - Mean of the binned pixels
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m-1
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-transp_nrt_l3-multi-4km_P1D	
Variables name in the NetCDF file	KD490
Long_name	Diffuse attenuation coefficient - Mean of the binned pixels
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m-1
Additional Information:	
Variables name in the NetCDF file	ZSD
Long_name	Secchi disk depth - Mean of the binned pixels
Standard_name:	secchi_depth_of_sea_water
Units	m
Additional Information:	

Variables name in the NetCDF file	SPM
Long_name	Inorganic suspended particulate matter in sea water - Mean of the binned pixels
Standard_name:	mass_concentration_of_suspended_matter_in_sea_water
Units	g/m3
Additional Information:	
Dataset: **cmems_obs-oc_glo_bgc-optics_nrt_l3-multi-4km_P1D**	
Variables name in the NetCDF file	CDM
Long_name	Coloured dissolved and detrital organic materials - Mean of the binned pixels
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_dissolved_organic_matter_and_non_algal_particles
Units	m-1
Additional Information:	
Variables name in the NetCDF file	BBP
Long_name	Particulate back-scattering coefficient at 443 nm - Mean of the binned pixels
Standard_name:	volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water_due_to_particles"
Units	m-1
Additional Information:	

Table 11 : list of the datasets and variable names and unit for the OCEANCOLOUR_GLO_BGC_L3_NRT_009_101 (Copernicus-GlobColour) product

II.3.1.2 OCEANCOLOUR_GLO_BGC_L4_NRT_009_102 (Copernicus-GlobColour)

OCEANCOLOUR_GLO_BGC_L4_NRT_009_102	
Dataset: **cmems_obs-oc_glo_bgc-reflectance_nrt_l4-olci-4km_P1M** **cmems_obs-oc_glo_bgc-reflectance_nrt_l4-olci-300m_P1M**	
Variables name in the NetCDF file	RRS400
Long_name	Fully normalised remote sensing reflectance at 400 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS412
Long_name	Fully normalised remote sensing reflectance at 412 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air

Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Fully normalised remote sensing reflectance at 443 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Fully normalised remote sensing reflectance at 490 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Fully normalised remote sensing reflectance at 510 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS560
Long_name	Fully normalised remote sensing reflectance at 560 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS620
Long_name	Fully normalised remote sensing reflectance at 620 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Fully normalised remote sensing reflectance at 665 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1

Additional Information:	
Variables name in the NetCDF file	RRS674
Long_name	Fully normalised remote sensing reflectance at 674 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS681
Long_name	Fully normalised remote sensing reflectance at 681 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS709
Long_name	Fully normalised remote sensing reflectance at 709 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-reflectance_nrt_l4-multi-4km_P1M	
Variables name in the NetCDF file	RRS412
Long_name	Fully normalised remote sensing reflectance at 412 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Fully normalised remote sensing reflectance at 443 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Fully normalised remote sensing reflectance at 490 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air

Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS555
Long_name	Fully normalised remote sensing reflectance at 555 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS670
Long_name	Fully normalised remote sensing reflectance at 670 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-plankton_nrt_l4-gapfree-multi-4km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll-a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	A mask variable identifies interpolated pixels from others.
Dataset: cmems_obs-oc_glo_bgc-plankton_nrt_l4-multi-4km_P1M	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll-a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	DIATO
Long_name	Diatoms Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_diatoms_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	DINO
Long_name	Dinophytes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_dinophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	GREEN

Long_name	Green Algae Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_greenalgae_and_prochlorophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	HAPTO
Long_name	Haptophytes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_haptophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PROCHLO
Long_name	Prochlorococcus Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_prochlorococcus_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PROKAR
Long_name	Prokaryotes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_prokaryotes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	MICRO
Long_name	Microphytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_microphytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	NANO
Long_name	Nanophytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_nanophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PICO
Long_name	Picophytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_picophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3

Additional Information:	
Dataset: **cmems_obs-oc_glo_bgc-plankton_nrt_l4-olci-4km_P1M** **cmems_obs-oc_glo_bgc-plankton_nrt_l4-olci-300m_P1M**	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Dataset: **cmems_obs-oc_glo_bgc-transp_nrt_l4-olci-4km_P1M**	
Variables name in the NetCDF file	KD490
Long_name	Diffuse attenuation coefficient - Mean of the binned pixels
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m-1
Additional Information:	
Dataset: **cmems_obs-oc_glo_bgc-transp_nrt_l4-gapfree-multi-4km_P1D**	
Variables name in the NetCDF file	KD490
Long_name	Diffuse attenuation coefficient - Mean of the binned pixels
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m-1
Additional Information:	A mask variable identifies interpolated pixels from others.
Variables name in the NetCDF file	ZSD
Long_name	Secchi disk depth - Mean of the binned pixels
Standard_name:	secchi_depth_of_sea_water
Units	m
Additional Information:	A mask variable identifies interpolated pixels from others.
Dataset: **cmems_obs-oc_glo_bgc-transp_nrt_l4-multi-4km_P1M**	
Variables name in the NetCDF file	KD490
Long_name	Diffuse attenuation coefficient - Mean of the binned pixels
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m-1
Additional Information:	
Variables name in the NetCDF file	ZSD
Long_name	Secchi disk depth - Mean of the binned pixels
Standard_name:	secchi_depth_of_sea_water
Units	m
Additional Information:	

Variables name in the NetCDF file	SPM
Long_name	Inorganic suspended particulate matter in sea water - Mean of the binned pixels
Standard_name:	mass_concentration_of_suspended_matter_in_sea_water
Units	g/m3
Additional Information:	
Dataset: **cmems_obs-oc_glo_bgc-optics_nrt_l4-multi-4km_P1M**	
Variables name in the NetCDF file	CDM
Long_name	Coloured dissolved and detrital organic materials - Mean of the binned pixels
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_dissolved_organic_matter_and_non_algal_particles
Units	m-1
Additional Information:	
Variables name in the NetCDF file	BBP
Long_name	Particulate back-scattering coefficient at 443 nm - Mean of the binned pixels
Standard_name:	volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water_due_to_particles"
Units	m-1
Additional Information:	
Dataset: **cmems_obs-oc_glo_bgc-pp_nrt_l4-multi-4km_P1M**	
Variables name in the NetCDF file	PP
Long_name	Primary Productivity – Mean of the binned pixels
Standard_name:	primary_productivity_of_biomass_expressed_as_carbon
Units	mg m-2 day-1
Additional Information:	

Table 12 : list of the datasets and variable names and unit for the OCEANCOLOUR_GLO_BGC_L4_NRT_009_102 (Copernicus-GlobColour) product

II.3.1.3 OCEANCOLOUR_GLO_BGC_L3_MY_009_103 (Copernicus-GlobColour)

OCEANCOLOUR_GLO_BGC_L3_MY_009_103	
Dataset: **Cmems_obs-oc_glo_bgc-reflectance_my_l3-olci-4km_P1D** **cmems_obs-oc_glo_bgc-reflectance_my_l3-olci-300m_P1D**	
Variables name in the NetCDF file	RRS400
Long_name	Fully normalised remote sensing reflectance at 400 nm – Mean of the binned pixels

Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS412
Long_name	Fully normalised remote sensing reflectance at 412 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Fully normalised remote sensing reflectance at 443 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Fully normalised remote sensing reflectance at 490 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Fully normalised remote sensing reflectance at 510 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS560
Long_name	Fully normalised remote sensing reflectance at 560 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS620
Long_name	Fully normalised remote sensing reflectance at 620 nm – Mean of the binned pixels

Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Fully normalised remote sensing reflectance at 665 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS674
Long_name	Fully normalised remote sensing reflectance at 674 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS681
Long_name	Fully normalised remote sensing reflectance at 681 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS709
Long_name	Fully normalised remote sensing reflectance at 709 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-reflectance_my_l3-multi-4km_P1D	
Variables name in the NetCDF file	RRS412
Long_name	Fully normalised remote sensing reflectance at 412 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS443

Long_name	Fully normalised remote sensing reflectance at 443 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Fully normalised remote sensing reflectance at 490 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS555
Long_name	Fully normalised remote sensing reflectance at 555 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS670
Long_name	Fully normalised remote sensing reflectance at 670 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-plankton_my_l3-olci-300m_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-plankton_my_l3-olci-4km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	CHL_gradient
Long_name	Norm of the horizontal gradient of the chlorophyll concentration

Standard_name:	magnitude_of_horizontal_gradient_of_mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3 km-1
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-plankton_my_l3-multi-4km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll-a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	DIATO
Long_name	Diatoms Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_diatoms_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	DINO
Long_name	Dinophytes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_dinophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	GREEN
Long_name	Green Algae Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_greenalgae_and_prochlorophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	HAPTO
Long_name	Haptophytes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_haptophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PROCHLO
Long_name	Prochlorococcus Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_prochlorococcus_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PROKAR
Long_name	Prokaryotes Chlorophyll a concentration - Mean of the binned pixels

Standard_name:	mass_concentration_of_prokaryotes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	MICRO
Long_name	Microphytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_microphytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	NANO
Long_name	Nanophytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_nanophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PICO
Long_name	Picophytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_picophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-transp_my_l3-olci-4km_P1D	
Variables name in the NetCDF file	KD490
Long_name	Diffuse attenuation coefficient - Mean of the binned pixels
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m-1
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-transp_my_l3-multi-4km_P1D	
Variables name in the NetCDF file	KD490
Long_name	Diffuse attenuation coefficient - Mean of the binned pixels
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m-1
Additional Information:	
Variables name in the NetCDF file	ZSD
Long_name	Secchi disk depth - Mean of the binned pixels

Standard_name:	secchi_depth_of_sea_water
Units	m
Additional Information:	
Variables name in the NetCDF file	SPM
Long_name	Inorganic suspended particulate matter in sea water - Mean of the binned pixels
Standard_name:	mass_concentration_of_suspended_matter_in_sea_water
Units	g/m3
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-optics_my_l3-multi-4km_P1D	
Variables name in the NetCDF file	CDM
Long_name	Coloured dissolved and detrital organic materials - Mean of the binned pixels
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_dissolved_organic_matter_and_non_algal_particles
Units	m-1
Additional Information:	
Variables name in the NetCDF file	BBP
Long_name	Particulate back-scattering coefficient at 443 nm - Mean of the binned pixels
Standard_name:	volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water_due_to_particles”
Units	m-1
Additional Information:	

Table 13 : list of the datasets and variable names and unit for the OCEANCOLOUR_GLO_BGC_L3_MY_009_103 (Copernicus-GlobColour) product

II.3.1.4 OCEANCOLOUR_GLO_BGC_L4_MY_009_104 (Copernicus-GlobColour)

OCEANCOLOUR_GLO_BGC_L4_MY_009_104	
Dataset: cmems_obs-oc_glo_bgc-reflectance_my_l4-olci-4km_P1M cmems_obs-oc_glo_bgc-reflectance_my_l4-olci-300m_P1M	
Variables name in the NetCDF file	RRS400
Long_name	Fully normalised remote sensing reflectance at 400 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	

Variables name in the NetCDF file	RRS412
Long_name	Fully normalised remote sensing reflectance at 412 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Fully normalised remote sensing reflectance at 443 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Fully normalised remote sensing reflectance at 490 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Fully normalised remote sensing reflectance at 510 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS560
Long_name	Fully normalised remote sensing reflectance at 560 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS620
Long_name	Fully normalised remote sensing reflectance at 620 nm – Mean of the binned pixels

Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Fully normalised remote sensing reflectance at 665 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS674
Long_name	Fully normalised remote sensing reflectance at 674 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS681
Long_name	Fully normalised remote sensing reflectance at 681 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS709
Long_name	Fully normalised remote sensing reflectance at 709 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-reflectance_my_l4-multi-4km_P1M	
Variables name in the NetCDF file	RRS412
Long_name	Fully normalised remote sensing reflectance at 412 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air

Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Fully normalised remote sensing reflectance at 443 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Fully normalised remote sensing reflectance at 490 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS555
Long_name	Fully normalised remote sensing reflectance at 555 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS670
Long_name	Fully normalised remote sensing reflectance at 670 nm – Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-plankton_my_l4-gapfree-multi-4km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll-a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	A mask variable identifies interpolated pixels from others.
Dataset:	

cmems_obs-oc_glo_bgc-plankton_my_l4-multi-4km_P1M	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll-a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	DIATO
Long_name	Diatoms Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_diatoms_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	DINO
Long_name	Dinophytes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_dinophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	GREEN
Long_name	Green Algae Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_greenalgae_and_prochlorophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	HAPTO
Long_name	Haptophytes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_haptophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PROCHLO
Long_name	Prochlorococcus Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_prochlorococcus_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PROKAR

Long_name	Prokaryotes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_prokaryotes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	MICRO
Long_name	Microphytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_microphytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	NANO
Long_name	Nanophytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_nanophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PICO
Long_name	Picophytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_picophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Dataset: **cmems_obs-oc_glo_bgc-plankton_my_l4-olci-4km_P1M** **cmems_obs-oc_glo_bgc-plankton_my_l4-olci-300m_P1M**	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Dataset: **cmems_obs-oc_glo_bgc-**transp**_my_l4-olci-4km_P1M**	
Variables name in the NetCDF file	KD490
Long_name	Diffuse attenuation coefficient - Mean of the binned pixels
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m-1

Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-transp_my_l4-gapfree-multi-4km_P1D	
Variables name in the NetCDF file	KD490
Long_name	Diffuse attenuation coefficient - Mean of the binned pixels
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m-1
Additional Information:	A mask variable identifies interpolated pixels from others.
Variables name in the NetCDF file	ZSD
Long_name	Secchi disk depth - Mean of the binned pixels
Standard_name:	secchi_depth_of_sea_water
Units	m
Additional Information:	A mask variable identifies interpolated pixels from others.
Dataset: cmems_obs-oc_glo_bgc-transp_my_l4-multi-4km_P1M	
Variables name in the NetCDF file	KD490
Long_name	Diffuse attenuation coefficient - Mean of the binned pixels
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m-1
Additional Information:	
Variables name in the NetCDF file	ZSD
Long_name	Secchi disk depth - Mean of the binned pixels
Standard_name:	secchi_depth_of_sea_water
Units	m
Additional Information:	
Variables name in the NetCDF file	SPM
Long_name	Inorganic suspended particulate matter in sea water - Mean of the binned pixels
Standard_name:	mass_concentration_of_suspended_matter_in_sea_water
Units	g/m3
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-optics_my_l4-multi-4km_P1M	
Variables name in the NetCDF file	CDM

Long_name	Coloured dissolved and detrital organic materials - Mean of the binned pixels
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_dissolved_organic_matter_and_non_algal_particles
Units	m-1
Additional Information:	
Variables name in the NetCDF file	BBP
Long_name	Particulate back-scattering coefficient at 443 nm - Mean of the binned pixels
Standard_name:	volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water_due_to_particles"
Units	m-1
Additional Information:	
Dataset: cmems_obs-oc_glo_bgc-pp_my_l4-multi-4km_P1M	
Variables name in the NetCDF file	PP
Long_name	Primary Productivity - Mean of the binned pixels
Standard_name:	primary_productivity_of_biomass_expressed_as_carbon
Units	mg m-2 day-1
Additional Information:	
Dataset cmems_obs-oc_glo_bgc-plankton_my_l4-multi-climatology-4km_P1D	
Variables name in the NetCDF file	CHL_mean
Long_name	Chlorophyll-a concentration - Mean of the binned pixels - Climatology mean
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	These daily climatology (336 files) have been created using the data falling into a moving temporal window of ± 5 days for the period 1998-2018
Variables name in the NetCDF file	CHL_standard_deviation
Long_name	Chlorophyll-a concentration - Mean of the binned pixels - Climatology standard_deviation
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	CHL_count
Long_name	Chlorophyll-a concentration - Mean of the binned pixels - Climatology count
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water number_of_observation

Units	N/A
Additional Information:	
Variables name in the NetCDF file	CHL_minimum
Long_name	Chlorophyll-a concentration - Mean of the binned pixels - Climatology minimum
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	CHL_maximum
Long_name	Chlorophyll-a concentration - Mean of the binned pixels – Climatology maximum
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	CHL_median
Long_name	Chlorophyll-a concentration - Mean of the binned pixels – Climatology median
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	CHL_percentile_standard_deviation
Long_name	Chlorophyll-a concentration - Mean of the binned pixels – Climatology percentile_standard_deviation
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	CHL_percentile_3
Long_name	Chlorophyll-a concentration - Mean of the binned pixels – Climatology percentile_3
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	CHL_percentile_97
Long_name	Chlorophyll-a concentration - Mean of the binned pixels – Climatology percentile_97
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	

Table 14 : list of the datasets and variable names and unit for the OCEANCOLOUR_GLO_BGC_L4_MY_009_104 (Copernicus-GlobColour) product

II.3.2 Global Ocean (OC-CCI)

II.3.2.1 OCEANCOLOUR_GLO_BGC_L3_MY_009_107

OCEANCOLOUR_GLO_BGC_L3_MY_009_107	
Dataset: c3s_obs-oc_glo_bgc-reflectance_my_l3-multi-4km_P1D	
Variables name in the NetCDF file	RRS412
Long_name	Multi-sensor Remote Sensing Reflectance at 412nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Multi-sensor Remote Sensing Reflectance at 443nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Multi-sensor Remote Sensing Reflectance at 490nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Multi-sensor Remote Sensing Reflectance at 510nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS560
Long_name	Multi-sensor Remote Sensing Reflectance at 560nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Multi-sensor Remote Sensing Reflectance at 665nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1

Additional Information:	
Dataset: c3s_obs-oc_glo_bgc-plankton_my_l3-multi-4km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	MICRO
Long_name	Multi-sensor Microphytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_microphytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	Also known as “Phytoplankton Size Classes” (PSCs)
Variables name in the NetCDF file	NANO
Long_name	Multi-sensor Nanophytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_nanophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	Also known as “Phytoplankton Size Classes” (PSCs)
Variables name in the NetCDF file	PICO
Long_name	Multi-sensor Picophytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_picophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	Also known as “Phytoplankton Size Classes” (PSCs)

Table 15 : list of the datasets and variable names and unit for the OCEANCOLOUR_GLO_BGC_L3_MY_009_107 (OC-CCI) product

II.3.2.2 OCEANCOLOUR_GLO_BGC_L4_MY_009_108

OCEANCOLOUR_GLO_BGC_L3_MY_009_108	
Dataset: c3s_obs-oc_glo_bgc-plankton_my_l4-multi-4km_P1M	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	

Table 16 : list of the datasets and variable names and unit for the OCEANCOLOUR_GLO_BGC_L3_MY_009_108 (OC-CCI) product

II.3.3 Atlantic

II.3.3.1 OCEANCOLOUR_ATL_BGC_L3_NRT_009_111

OCEANCOLOUR_ATL_BGC_L3_NRT_009_111	
Dataset: cmems_obs-oc_atl_bgc-reflectance_nrt_l3-olci-300m_P1D	
Variables name in the NetCDF file	RRS400
Long_name	Fully normalised remote sensing reflectance at 400 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS412
Long_name	Fully normalised remote sensing reflectance at 412 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Fully normalised remote sensing reflectance at 443 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Fully normalised remote sensing reflectance at 490 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Fully normalised remote sensing reflectance at 510 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS560

Long_name	Fully normalised remote sensing reflectance at 560 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS620
Long_name	Fully normalised remote sensing reflectance at 620 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Fully normalised remote sensing reflectance at 665 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS674
Long_name	Fully normalised remote sensing reflectance at 674 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS681
Long_name	Fully normalised remote sensing reflectance at 681 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS709
Long_name	Fully normalised remote sensing reflectance at 709 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Dataset: cmems_obs-oc_atl_bgc-reflectance_nrt_l3-multi-1km_P1D	
Variables name in the NetCDF file	RRS412

Long_name	Fully normalised remote sensing reflectance at 412 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Fully normalised remote sensing reflectance at 443 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Fully normalised remote sensing reflectance at 490 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS555
Long_name	Fully normalised remote sensing reflectance at 555 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS670
Long_name	Fully normalised remote sensing reflectance at 670 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Dataset: cmems_obs-oc_atl_bgc-plankton_nrt_l3-olci-300m_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Dataset: cmems_obs-oc_atl_bgc-plankton_nrt_l3-olci-1km_P1D	
Variables name in the NetCDF file	CHL

Long_name	Chlorophyll concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	CHL_gradient
Long_name	Norm of the horizontal gradient of the chlorophyll concentration
Standard_name:	magnitude_of_horizontal_gradient_of_mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3 km-1
Additional Information:	
Dataset: cmems_obs-oc_atl_bgc-plankton_nrt_l3-multi-1km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll-a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	DIATO
Long_name	Diatoms Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_diatoms_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	DINO
Long_name	Dinophytes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_dinophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	GREEN
Long_name	Green Algae Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_greenalgae_and_prochlorophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	HAPTO
Long_name	Haptophytes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_haptophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PROCHLO

Long_name	Prochlorococcus Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_prochlorococcus_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PROKAR
Long_name	Prokaryotes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_prokaryotes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	MICRO
Long_name	Microphytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_microphytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	NANO
Long_name	Nanophytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_nanophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PICO
Long_name	Picophytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_picophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Dataset: cmems_obs-oc_atl_bgc-transp_nrt_l3-multi-1km_P1D	
Variables name in the NetCDF file	SPM
Long_name	Inorganic suspended particulate matter in sea water - Mean of the binned pixels
Standard_name:	mass_concentration_of_suspended_matter_in_sea_water
Units	g/m3
Additional Information:	
Variables name in the NetCDF file	KD490
Long_name	Diffuse attenuation coefficient - Mean of the binned pixels

Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m-1
Additional Information:	
Variables name in the NetCDF file	ZSD
Long_name	Secchi disk depth - Mean of the binned pixels
Standard_name:	secchi_depth_of_sea_water
Units	m
Additional Information:	
Dataset: cmems_obs-oc_atl_bgc-optics_nrt_l3-multi-1km_P1D	
Variables name in the NetCDF file	CDM
Long_name	Coloured dissolved and detrital organic materials - Mean of the binned pixels
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_dissolved_organic_matter_and_non_algal_particles
Units	m-1
Additional Information:	
Variables name in the NetCDF file	BBP
Long_name	Particulate back-scattering coefficient at 443 nm - Mean of the binned pixels
Standard_name:	volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water_due_to_particles
Units	m-1
Additional Information:	

Table 17 : list of the datasets and variable names and unit for the OCEANCOLOUR_ATL_BGC_L3_NRT_009_111 (OC-CCI) product

II.3.3.2 OCEANCOLOUR_ATL_BGC_L3_MY_009_113

OCEANCOLOUR_ATL_BGC_L3_MY_009_113	
Dataset: cmems_obs-oc_atl_bgc-reflectance_my_l3-olci-300m_P1D	
Variables name in the NetCDF file	RRS400
Long_name	Fully normalised remote sensing reflectance at 400 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS412
Long_name	Fully normalised remote sensing reflectance at 412 nm - Mean of the binned pixels

Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to _downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Fully normalised remote sensing reflectance at 443 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to _downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Fully normalised remote sensing reflectance at 490 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to _downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Fully normalised remote sensing reflectance at 510 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to _downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS560
Long_name	Fully normalised remote sensing reflectance at 560 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to _downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS620
Long_name	Fully normalised remote sensing reflectance at 620 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to _downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Fully normalised remote sensing reflectance at 665 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to _downwelling_radiative_flux_in_air

Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS674
Long_name	Fully normalised remote sensing reflectance at 674 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS681
Long_name	Fully normalised remote sensing reflectance at 681 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS709
Long_name	Fully normalised remote sensing reflectance at 709 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Dataset: cmems_obs-oc_atl_bgc-reflectance_my_l3-multi-1km_P1D	
Variables name in the NetCDF file	RRS412
Long_name	Fully normalised remote sensing reflectance at 412 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Fully normalised remote sensing reflectance at 443 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Fully normalised remote sensing reflectance at 490 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air

Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS555
Long_name	Fully normalised remote sensing reflectance at 555 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Variables name in the NetCDF file	RRS670
Long_name	Fully normalised remote sensing reflectance at 670 nm - Mean of the binned pixels
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr-1
Additional Information:	
Dataset: cmems_obs-oc_atl_bgc-plankton_my_l3-olci-300m_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Dataset: cmems_obs-oc_atl_bgc-plankton_my_l3-olci-1km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	CHL_gradient
Long_name	Norm of the horizontal gradient of the chlorophyll concentration
Standard_name:	magnitude_of_horizontal_gradient_of_mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3 km-1
Additional Information:	
Dataset: cmems_obs-oc_atl_bgc-plankton_my_l3-multi-1km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll-a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	DIATO

Long_name	Multi-sensor Diatom Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_diatoms_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	DINO
Long_name	Multi-sensor Dinophyte Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_dinophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	GREEN
Long_name	Green Algae Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_greenalgae_and_prochlorophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	HAPTO
Long_name	Haptophytes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_haptophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PROCHLO
Long_name	Prochlorococcus Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_prochlorococcus_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PROKAR
Long_name	Prokaryotes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_prokaryotes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	MICRO
Long_name	Multi-sensor Microphytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_microphytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3

Additional Information:	Also known as “Phytoplankton Size Classes” (PSCs)
Variables name in the NetCDF file	NANO
Long_name	Multi-sensor Nanophytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_nanophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	Also known as “Phytoplankton Size Classes” (PSCs)
Variables name in the NetCDF file	PICO
Long_name	Multi-sensor Picophytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_picophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	Also known as “Phytoplankton Size Classes” (PSCs)
Dataset: cmems_obs-oc_atl_bgc-transp_my_l3-multi-1km_P1D	
Variables name in the NetCDF file	SPM
Long_name	Inorganic suspended particulate matter in sea water - Mean of the binned pixels
Standard_name:	mass_concentration_of_suspended_matter_in_sea_water
Units	g/m3
Additional Information:	
Variables name in the NetCDF file	KD490
Long_name	Diffuse attenuation coefficient - Mean of the binned pixels
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m-1
Additional Information:	
Variables name in the NetCDF file	ZSD
Long_name	Secchi disk depth - Mean of the binned pixels
Standard_name:	secchi_depth_of_sea_water
Units	m
Additional Information:	
Dataset: cmems_obs-oc_atl_bgc-optics_my_l3-multi-1km_P1D	
Variables name in the NetCDF file	CDM
Long_name	Coloured dissolved and detrital organic materials - Mean of the binned pixels
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_dissolved_organic_matter_and_non_algal_particles
Units	m-1
Additional Information:	
Variables name in the NetCDF file	BBP

Long_name	Particulate back-scattering coefficient at 443 nm - Mean of the binned pixels
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_particles
Units	m-1
Additional Information:	

Table 18 : list of the datasets and variable names and unit for the OCEANCOLOUR_ATL_BGC_L3_MY_009_113 product

II.3.3.3 OCEANCOLOUR_ATL_BGC_L4_NRT_009_116 (Copernicus-GlobColour)

OCEANCOLOUR_ATL_BGC_L4_NRT_009_116	
Dataset: cmems_obs-oc_atl_bgc-plankton_nrt_l4-gapfree-multi-1km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll-a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	A mask variable identifies interpolated pixels from others.
Dataset: cmems_obs-oc_atl_bgc-plankton_nrt_l4-multi-1km_P1M	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll-a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	DIATO
Long_name	Multi-sensor Diatom Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_diatoms_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	DINO
Long_name	Multi-sensor Dinophyte Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_dinophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	GREEN
Long_name	Green Algae Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_greenalgae_and_prochlorophytes_expressed_as_chlorophyll_in_sea_water

Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	HAPTO
Long_name	Haptophytes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_haptophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PROCHLO
Long_name	Prochlorococcus Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_prochlorococcus_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PROKAR
Long_name	Prokaryotes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_prokaryotes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	MICRO
Long_name	Multi-sensor Microphytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_microphytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	Also known as “Phytoplankton Size Classes” (PSCs)
Variables name in the NetCDF file	NANO
Long_name	Multi-sensor Nanophytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_nanophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	Also known as “Phytoplankton Size Classes” (PSCs)
Variables name in the NetCDF file	PICO
Long_name	Multi-sensor Picophytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_picophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	Also known as “Phytoplankton Size Classes” (PSCs)
Dataset: **cmems_obs-oc_atl_bgc-pp_nrt_l4-multi-1km_P1M**	

Variables name in the NetCDF file	PP
Long_name	Primary Productivity – Mean of the binned pixels
Standard_name:	primary_productivity_of_biomass_expressed_as_carbon
Units	mg m-2 day-1
Additional Information:	

Table 19 : list of the datasets and variable names and unit for the OCEANCOLOUR_ATL_BGC_L4_NRT_009_116 (Copernicus-GlobColour) product

II.3.3.4 OCEANCOLOUR_ATL_BGC_L4_MY_009_118 (Copernicus-GlobColour)

OCEANCOLOUR_ATL_BGC_L4_MY_009_118	
Dataset: cmems_obs-oc_atl_bgc-plankton_my_l4-gapfree-multi-1km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll-a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	A mask variable identifies interpolated pixels from others.
Dataset: cmems_obs-oc_atl_bgc-plankton_my_l4-multi-1km_P1M	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll-a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m-3
Additional Information:	
Variables name in the NetCDF file	DIATO
Long_name	Multi-sensor Diatom Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_diatoms_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	DINO
Long_name	Multi-sensor Dinophyte Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_dinophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	GREEN
Long_name	Green Algae Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_greenalgae_and_prochlorophytes_expressed_as_chlorophyll_in_sea_water

Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	HAPTO
Long_name	Haptophytes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_haptophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PROCHLO
Long_name	Prochlorococcus Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_prochlorococcus_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	PROKAR
Long_name	Prokaryotes Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_prokaryotes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	
Variables name in the NetCDF file	MICRO
Long_name	Multi-sensor Microphytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_microphytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	Also known as “Phytoplankton Size Classes” (PSCs)
Variables name in the NetCDF file	NANO
Long_name	Multi-sensor Nanophytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_nanophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	Also known as “Phytoplankton Size Classes” (PSCs)
Variables name in the NetCDF file	PICO
Long_name	Multi-sensor Picophytoplankton Chlorophyll a concentration - Mean of the binned pixels
Standard_name:	mass_concentration_of_picophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	Also known as “Phytoplankton Size Classes” (PSCs)
Dataset: cmems_obs-oc_atl_bgc-pp_my_l4-multi-1km_P1M	

Variables name in the NetCDF file	PP
Long_name	Primary Productivity – Mean of the binned pixels
Standard_name:	primary_productivity_of_biomass_expressed_as_carbon
Units	mg m-2 day-1
Additional Information:	

Table 20 : list of the datasets and variable names and unit for the OCEANCOLOUR_ATL_BGC_L4_MY_009_118 (Copernicus-GlobColour) product

II.3.4 Arctic Region

II.3.4.1 OCEANCOLOUR_ARC_BGC_L3_NRT_009_121

OCEANCOLOUR_ARC_BGC_L3_NRT_009_121	
Dataset: cmems_obs-oc_arc_bgc-reflectance_nrt_l3-olci-300m_P1D	
Variables name in the NetCDF file	RRS400
Long_name	Remote Sensing Reflectance at rrs400
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS412_5
Long_name	Remote Sensing Reflectance at rrs412_5
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS442_5
Long_name	Remote Sensing Reflectance at rrs442_5
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Remote Sensing Reflectance at rrs490
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Remote Sensing Reflectance at rrs510
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1

Additional Information:	
Variables name in the NetCDF file	RRS560
Long_name	Remote Sensing Reflectance at rrs560
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS620
Long_name	Remote Sensing Reflectance at rrs620
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Remote Sensing Reflectance at rrs665
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS673_75
Long_name	Remote Sensing Reflectance at rrs673_75
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS681_25
Long_name	Remote Sensing Reflectance at rrs681_25
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS708_75
Long_name	Remote Sensing Reflectance at rrs708_75
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Dataset: cmems_obs-oc_arc_bgc-plankton_nrt_l3-olci-300m_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	

Dataset: cmems_obs-oc_arc_bgc-transp_nrt_l3-olci-300m_P1D	
Variables name in the NetCDF file	KD490
Long_name	OLCI Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	

Table 21 : list of the datasets and variable names and unit for the OCEANCOLOUR_ARC_BGC_L3_NRT_009_121 product

II.3.4.2 OCEANCOLOUR_ARC_BGC_L4_NRT_009_122

OCEANCOLOUR_ARC_BGC_L4_NRT_009_122	
Dataset: cmems_obs-oc_arc_bgc-plankton_nrt_l4-olci-300m_P1M	
Variables name in the NetCDF file	CHL
Long_name	OLCI Monthly Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	
Dataset: cmems_obs-oc_arc_bgc-transp_nrt_l4-olci-300m_P1M	
Variables name in the NetCDF file	KD490
Long_name	OLCI Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	

Table 22 : list of the datasets and variable names and unit for the OCEANCOLOUR_ARC_BGC_L4_NRT_009_122 product

II.3.4.3 OCEANCOLOUR_ARC_BGC_L3_MY_009_123

OCEANCOLOUR_ARC_BGC_L3_MY_009_123	
Dataset: cmems_obs-oc_arc_bgc-reflectance_my_l3-olci-300m_P1D	
Variables name in the NetCDF file	RRS400
Long_name	Remote Sensing Reflectance at rrs400
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS412_5
Long_name	Remote Sensing Reflectance at rrs412_5

Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS442_5
Long_name	Remote Sensing Reflectance at rrs442_5
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Remote Sensing Reflectance at rrs490
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Remote Sensing Reflectance at rrs510
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS560
Long_name	Remote Sensing Reflectance at rrs560
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS620
Long_name	Remote Sensing Reflectance at rrs620
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Remote Sensing Reflectance at rrs665
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS673_75
Long_name	Remote Sensing Reflectance at rrs673_75
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air

Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS681_25
Long_name	Remote Sensing Reflectance at rrs681_25
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS708_75
Long_name	Remote Sensing Reflectance at rrs708_75
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Dataset: cmems_obs-oc_arc_bgc-reflectance_my_l3-multi-4km_P1D	
Variables name in the NetCDF file	RRS412
Long_name	Multi-sensor Remote Sensing Reflectance at 412nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Multi-sensor Remote Sensing Reflectance at 443nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Multi-sensor Remote Sensing Reflectance at 490nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Multi-sensor Remote Sensing Reflectance at 510nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS560
Long_name	Multi-sensor Remote Sensing Reflectance at 560nm

Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Multi-sensor Remote Sensing Reflectance at 665nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Dataset: cmems_obs-oc_arc_bgc-plankton_my_l3-olci-300m_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	
Dataset: cmems_obs-oc_arc_bgc-plankton_my_l3-multi-4km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	
Dataset: cmems_obs-oc_arc_bgc-transp_my_l3-olci-300m_P1D	
Variables name in the NetCDF file	KD490
Long_name	OLCI Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	
Dataset: cmems_obs-oc_arc_bgc-transp_my_l3-multi-4km_P1D	
Variables name in the NetCDF file	KD490
Long_name	Multi-sensor Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	

Table 23 : list of the datasets and variable names and unit for the OCEANCOLOUR_ARC_BGC_L3_MY_009_123 product

II.3.4.4 OCEANCOLOUR_ARC_BGC_L4_MY_009_124

OCEANCOLOUR_ARC_BGC_L4_MY_009_124	
Dataset: cmems_obs-oc_arc_bgc-plankton_my_l4-olci-300m_P1M	
Variables name in the NetCDF file	CHL
Long_name	OLCI Monthly Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	
Dataset: cmems_obs-oc_arc_bgc-plankton_my_l4-multi-4km_P1M	
Variables name in the NetCDF file	CHL
Long_name	Multi-sensor Monthly Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	
Dataset: cmems_obs-oc_arc_bgc-plankton_my_l4-multi-4km_P1M	
Variables name in the NetCDF file	CHL
Long_name	Climatology - Mean values of Chlorophyl Concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	These climatology maps have been created using the data falling into a weighted spatial window of 3x3 pixels and a moving temporal window of ± 5 days.

Table 24 : list of the datasets and variable names and unit for the OCEANCOLOUR_ARC_BGC_L4_MY_009_124 product

II.3.5 Baltic Sea

II.3.5.1 OCEANCOLOUR_BAL_BGC_L3_NRT_009_131

OCEANCOLOUR_BAL_BGC_L3_NRT_009_131	
Dataset: cmems_obs-oc_bal_bgc-reflectance_nrt_l3-olci-300m_P1D	
Variables name in the NetCDF file	RRS400
Long_name	Remote Sensing Reflectance at rrs400
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS412_5
Long_name	Remote Sensing Reflectance at rrs412_5

Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS442_5
Long_name	Remote Sensing Reflectance at rrs442_5
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Remote Sensing Reflectance at rrs490
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Remote Sensing Reflectance at rrs510
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS560
Long_name	Remote Sensing Reflectance at rrs560
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS620
Long_name	Remote Sensing Reflectance at rrs620
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Remote Sensing Reflectance at rrs665
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS673_75
Long_name	Remote Sensing Reflectance at rrs673_75
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air

Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS681_25
Long_name	Remote Sensing Reflectance at rrs681_25
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS708_75
Long_name	Remote Sensing Reflectance at rrs708_75
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Dataset: cmems_obs-oc_bal_bgc-plankton_nrt_l3-olci-300m_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained from OLCI data with Neural Network. ([RD11],[RD12],[RD13],[RD14])
Variables name in the NetCDF file	CRYPTO
Long_name	Multi-sensor Cryptophyte Chlorophyll a concentration
Standard_name:	mass_concentration_of_cryptophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation.
Variables name in the NetCDF file	DIATO
Long_name	Multi-sensor Diatom Chlorophyll a concentration
Standard_name:	mass_concentration_of_diatoms_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation.
Variables name in the NetCDF file	DINO
Long_name	Multi-sensor Dinophyte Chlorophyll a concentration
Standard_name:	mass_concentration_of_dinophytes_expressed_as_chlorophyll_in_sea_water

Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation.
Variables name in the NetCDF file	GREEN
Long_name	Multi-sensor Green algae and Prochlorophyte Chlorophyll a concentration
Standard_name:	mass_concentration_of_greenalgae_and_prochlorophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation.
Variables name in the NetCDF file	PROKAR
Long_name	Multi-sensor Prokaryote Chlorophyll a concentration
Standard_name:	mass_concentration_of_prokaryotes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation.
Variables name in the NetCDF file	MICRO
Long_name	Multi-sensor Microphytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_microphytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	MICRO consists of DIATO and DINO
Variables name in the NetCDF file	NANO
Long_name	Multi-sensor Nanophytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_nanophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	NANO consists of CRYPTO and half of the GREEN group
Variables name in the NetCDF file	PICO
Long_name	Multi-sensor Picophytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_picophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	PICO consists of half GREEN and PROKAR
Dataset: cmems_obs-oc_bal_bgc-transp_nrt_l3-olci-300m_P1D	

Variables name in the NetCDF file	KD490
Long_name	OLCI Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	
Dataset: cmems_obs-oc_bal_bgc-optics_nrt_l3-olci-300m_P1D	
Variables name in the NetCDF file	ADG443
Long_name	Absorption due to gelbstoff and detrital material at 443 nm, QAA algorithm
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_dissolved_organic_matter_and_non_algal_particles
Units	m^-1
Additional Information:	
Variables name in the NetCDF file	APH443
Long_name	Absorption due to phytoplankton at 443 nm, QAA algorithm
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_phytoplankton
Units	m^-1
Additional Information:	
Variables name in the NetCDF file	BBP443
Long_name	Particulate back-scattering coefficient at 443 nm, QAA algorithm
Standard_name:	volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water_due_to_particles
Units	m^-1
Additional Information:	

Table 25 : list of the datasets and variable names and unit for the OCEANCOLOUR_BAL_BGC_L3_NRT_009_131 product

II.3.5.2 OCEANCOLOUR_BAL_BGC_L4_NRT_009_132

OCEANCOLOUR_BAL_BGC_L4_NRT_009_132	
Dataset: cmems_obs-oc_bal_bgc-plankton_nrt_l4-olci-300m_P1M	
Variables name in the NetCDF file	CHL
Long_name	OLCI Monthly Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained from OLCI data with Neural Network. ([RD12],[RD13],[RD14])

Table 26 : list of the datasets and variable names and unit for the OCEANCOLOUR_BAL_BGC_L4_NRT_009_132 product

II.3.5.3 OCEANCOLOUR_BAL_BGC_L3_MY_009_133

OCEANCOLOUR_BAL_BGC_L3_MY_009_133	
Dataset: cmems_obs-oc_bal_bgc-reflectance_my_l3-olci-300m_P1D	
Variables name in the NetCDF file	RRS400
Long_name	Remote Sensing Reflectance at rrs400
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS412_5
Long_name	Remote Sensing Reflectance at rrs412_5
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS442_5
Long_name	Remote Sensing Reflectance at rrs442_5
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Remote Sensing Reflectance at rrs490
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Remote Sensing Reflectance at rrs510
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS560
Long_name	Remote Sensing Reflectance at rrs560
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS620
Long_name	Remote Sensing Reflectance at rrs620

Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Remote Sensing Reflectance at rrs665
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS673_75
Long_name	Remote Sensing Reflectance at rrs673_75
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS681_25
Long_name	Remote Sensing Reflectance at rrs681_25
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS708_75
Long_name	Remote Sensing Reflectance at rrs708_75
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Dataset: cmems_obs-oc_bal_bgc-reflectance_my_l3-multi-1km_P1D	
Variables name in the NetCDF file	RRS412
Long_name	Multi-sensor Remote Sensing Reflectance at 412nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Multi-sensor Remote Sensing Reflectance at 443nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Multi-sensor Remote Sensing Reflectance at 490nm

Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Multi-sensor Remote Sensing Reflectance at 510nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS560
Long_name	Multi-sensor Remote Sensing Reflectance at 560nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Multi-sensor Remote Sensing Reflectance at 665nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Dataset: cmems_obs-oc_bal_bgc-plankton_my_l3-olci-300m_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained from OLCI data with Neural Network. ([RD11],[RD12],[RD13],[RD14])
Dataset: cmems_obs-oc_bal_bgc-plankton_my_l3-multi-1km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained from multi-sensor data with Neural Network. ([RD11],[RD12],[RD13],[RD14])
Variables name in the NetCDF file	CRYPTO
Long_name	Multi-sensor Cryptophyte Chlorophyll a concentration
Standard_name:	mass_concentration_of_cryptophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3

Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation.
Variables name in the NetCDF file	DIATO
Long_name	Multi-sensor Diatom Chlorophyll a concentration
Standard_name:	mass_concentration_of_diatoms_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation.
Variables name in the NetCDF file	DINO
Long_name	Multi-sensor Dinophyte Chlorophyll a concentration
Standard_name:	mass_concentration_of_dinophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation.
Variables name in the NetCDF file	GREEN
Long_name	Multi-sensor Green algae and Prochlorophyte Chlorophyll a concentration
Standard_name:	mass_concentration_of_greenalgae_and_prochlorophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation.
Variables name in the NetCDF file	PROKAR
Long_name	Multi-sensor Prokaryote Chlorophyll a concentration
Standard_name:	mass_concentration_of_prokaryotes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation.
Variables name in the NetCDF file	MICRO
Long_name	Multi-sensor Microphytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_microphytoplankton_expressed_as_chlorophyll_in_sea_water

Units	milligram m^-3
Additional Information:	MICRO consists of DIATO and DINO
Variables name in the NetCDF file	NANO
Long_name	Multi-sensor Nanophytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_nanophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	NANO consists of CRYPTO and half of the GREEN group
Variables name in the NetCDF file	PICO
Long_name	Multi-sensor Picophytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_picophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	PICO consists of half GREEN and PROKAR
Dataset: cmems_obs-oc_bal_bgc-transp_my_l3-olci-300m_P1D	
Variables name in the NetCDF file	KD490
Long_name	OLCI Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	
Dataset: cmems_obs-oc_bal_bgc-transp_my_l3-multi-1km_P1D	
Variables name in the NetCDF file	KD490
Long_name	Multi-sensor Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	
Dataset: cmems_obs-oc_bal_bgc-optics_my_l3-olci-300m_P1D	
Variables name in the NetCDF file	ADG443
Long_name	Absorption due to gelbstoff and detrital material at 443 nm, QAA algorithm
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_dissolved_organic_matter_and_non_algal_particles
Units	m^-1
Additional Information:	
Variables name in the NetCDF file	APH443
Long_name	Absorption due to phytoplankton at 443 nm, QAA algorithm
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_phytoplankton
Units	m^-1

Additional Information:	
Variables name in the NetCDF file	BBP443
Long_name	Particulate back-scattering coefficient at 443 nm, QAA algorithm
Standard_name:	volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water_due_to_particles
Units	m^-1
Additional Information:	

Table 27 : list of the datasets and variable names and unit for the OCEANCOLOUR_BAL_BGC_L3_MY_009_133 product

II.3.5.4 OCEANCOLOUR_BAL_BGC_L4_MY_009_134

OCEANCOLOUR_BAL_BGC_L4_MY_009_134	
Dataset: cmems_obs-oc_bal_bgc-plankton_my_l4-olci-300m_P1M	
Variables name in the NetCDF file	CHL
Long_name	OLCI Monthly Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained from OLCI data with Neural Network. ([RD12],[RD13],[RD14])
Dataset: cmems_obs-oc_bal_bgc-plankton_my_l4-multi-1km_P1M	
Variables name in the NetCDF file	CHL
Long_name	Multi-sensor Monthly Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained from multi-sensor data with Neural Network. ([RD12],[RD13],[RD14])
Dataset: cmems_obs-oc_bal_bgc-pp_my_l4-multi-4km_P1D	
Variables name in the NetCDF file	PP
Long_name	Multi-sensor Integrated Primary Production
Standard_name:	primary_productivity_of_biomass_expressed_as_carbon
Units	g m^-2 day^-1
Additional Information:	
Dataset: cmems_obs-oc_bal_bgc-pp_my_l4-multi-4km_P1M	
Variables name in the NetCDF file	PP
Long_name	Monthly Multi-sensor Integrated Primary Production
Standard_name:	primary_productivity_of_biomass_expressed_as_carbon
Units	g m^-2 day^-1
Additional Information:	

Table 28 : list of the datasets and variable names and unit for the OCEANCOLOUR_BAL_BGC_L4_MY_009_134 product

II.3.6 Mediterranean Sea

II.3.6.1 OCEANCOLOUR_MED_BGC_L3_NRT_009_141

OCEANCOLOUR_MED_BGC_L3_NRT_009_141	
Dataset: cmems_obs-oc_med_bgc-reflectance_nrt_l3-olci-300m_P1D	
Variables name in the NetCDF file	RRS400
Long_name	Remote Sensing Reflectance at rrs400
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS412_5
Long_name	Remote Sensing Reflectance at rrs412_5
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS442_5
Long_name	Remote Sensing Reflectance at rrs442_5
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Remote Sensing Reflectance at rrs490
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Remote Sensing Reflectance at rrs510
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS560
Long_name	Remote Sensing Reflectance at rrs560
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS620

Long_name	Remote Sensing Reflectance at rrs620
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Remote Sensing Reflectance at rrs665
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS673_75
Long_name	Remote Sensing Reflectance at rrs673_75
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS681_25
Long_name	Remote Sensing Reflectance at rrs681_25
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS708_75
Long_name	Remote Sensing Reflectance at rrs708_75
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Dataset: cmems_obs-oc_med_bgc-reflectance_nrt_l3-multi-1km_P1D	
Variables name in the NetCDF file	RRS412
Long_name	Multi-sensor Remote Sensing Reflectance at 412nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Multi-sensor Remote Sensing Reflectance at 443nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS490

Long_name	Multi-sensor Remote Sensing Reflectance at 490nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Multi-sensor Remote Sensing Reflectance at 510nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS555
Long_name	Multi-sensor Remote Sensing Reflectance at 555nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS670
Long_name	Multi-sensor Remote Sensing Reflectance at 670nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Dataset: cmems_obs-oc_med_bgc-plankton_nrt_l3-olci-300m_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD1],[RD2],[RD3],[RD6])
Dataset: cmems_obs-oc_med_bgc-plankton_nrt_l3-multi-1km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD1],[RD2],[RD3],[RD6])
Variables name in the NetCDF file	CRYPTO
Long_name	Multi-sensor Cryptophyte Chlorophyll a concentration

Standard_name:	mass_concentration_of_cryptophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation ([RD7])
Variables name in the NetCDF file	DIATO
Long_name	Multi-sensor Diatom Chlorophyll a concentration
Standard_name:	mass_concentration_of_diatoms_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation ([RD7])
Variables name in the NetCDF file	DINO
Long_name	Multi-sensor Dinophyte Chlorophyll a concentration
Standard_name:	mass_concentration_of_dinophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation ([RD7])
Variables name in the NetCDF file	GREEN
Long_name	Multi-sensor Green algae and Prochlorophyte Chlorophyll a concentration
Standard_name:	mass_concentration_of_greenalgae_and_prochlorophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation ([RD7])
Variables name in the NetCDF file	HAPTO
Long_name	Multi-sensor Haptophyte Chlorophyll a concentration
Standard_name:	mass_concentration_of_haptophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation ([RD7])
Variables name in the NetCDF file	PROKAR

Long_name	Multi-sensor Prokaryote Chlorophyll a concentration
Standard_name:	mass_concentration_of_prokaryotes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation ([RD7]).
Variables name in the NetCDF file	MICRO
Long_name	Multi-sensor Microphytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_microphytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	MICRO consists of DIATO and DINO
Variables name in the NetCDF file	NANO
Long_name	Multi-sensor Nanophytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_nanophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	NANO consists of CRYPTO and HAPTO
Variables name in the NetCDF file	PICO
Long_name	Multi-sensor Picophytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_picophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	PICO consists of GREEN and PROKAR. In the ultra-oligotrophic condition HAPTO were divided between NANO and PICO following the adjustment of [RD15]
Dataset: cmems_obs-oc_med_bgc-transp_nrt_l3-olci-300m_P1D	
Variables name in the NetCDF file	KD490
Long_name	OLCI Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	
Dataset: cmems_obs-oc_med_bgc-transp_nrt_l3-multi-1km_P1D	
Variables name in the NetCDF file	KD490
Long_name	Multi-sensor Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	

Dataset: cmems_obs-oc_med_bgc-optics_nrt_l3-multi-1km_P1D	
Variables name in the NetCDF file	ADG443
Long_name	Absorption due to gelbstoff and detrital material at 443 nm, QAA algorithm
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_dissolved_organic_matter_and_non_algal_particles
Units	m^-1
Additional Information:	
Variables name in the NetCDF file	APH443
Long_name	Absorption due to phytoplankton at 443 nm, QAA algorithm
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_phytoplankton
Units	m^-1
Additional Information:	
Variables name in the NetCDF file	BBP443
Long_name	Particulate back-scattering coefficient at 443 nm, QAA algorithm
Standard_name:	volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water_due_to_particles
Units	m^-1
Additional Information:	

Table 29 : list of the datasets and variable names and unit for the OCEANCOLOUR_MED_BGC_L3_NRT_009_141 product

II.3.6.2 OCEANCOLOUR_MED_BGC_L4_NRT_009_142

OCEANCOLOUR_MED_BGC_L4_NRT_009_142	
Dataset: cmems_obs-oc_med_bgc-plankton_nrt_l4-olci-300m_P1M	
Variables name in the NetCDF file	CHL
Long_name	OLCI Monthly Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD1],[RD2],[RD3],[RD6])
Dataset: cmems_obs-oc_med_bgc-plankton_nrt_l4-multi-1km_P1M	
Variables name in the NetCDF file	CHL
Long_name	Monthly Multi-sensor and multi water-type Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3

Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD1],[RD2],[RD3],[RD6])
Dataset: cmems_obs-oc_med_bgc-plankton_nrt_l4-gapfree-multi-1km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Multi-sensor, multi water-type, interpolated Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD1],[RD2],[RD3],[RD6]). The interpolated gap-free Chl concentration is estimated by means of a modified version of the DINEOF algorithm by GOS ([RD4],[RD10]).
Dataset: cmems_obs-oc_med_bgc-transp_nrt_l4-olci-300m_P1M	
Variables name in the NetCDF file	KD490
Long_name	OLCI Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	
Dataset: cmems_obs-oc_med_bgc-transp_nrt_l4-multi-1km_P1M	
Variables name in the NetCDF file	KD490
Long_name	Monthly Multi-sensor Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	
Dataset: cmems_obs-oc_med_bgc-pp_nrt_l4-multi-4km_P1D	
Variables name in the NetCDF file	PP
Long_name	Multi-sensor Integrated Primary Production
Standard_name:	primary_productivity_of_biomass_expressed_as_carbon
Units	g m^-2 day^-1
Additional Information:	
Dataset: cmems_obs-oc_med_bgc-pp_nrt_l4-multi-4km_P1M	
Variables name in the NetCDF file	PP
Long_name	Monthly Multi-sensor Integrated Primary Production
Standard_name:	primary_productivity_of_biomass_expressed_as_carbon
Units	g m^-2 day^-1

Additional Information:	

Table 30 : list of the datasets and variable names and unit for the OCEANCOLOUR_MED_BGC_L4_NRT_009_142 product

II.3.6.3 OCEANCOLOUR_MED_BGC_L3_MY_009_143

OCEANCOLOUR_MED_BGC_L3_MY_009_143	
Dataset: cmems_obs-oc_med_bgc-reflectance_my_l3-olci-300m_P1D	
Variables name in the NetCDF file	RRS400
Long_name	Remote Sensing Reflectance at rrs400
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS412_5
Long_name	Remote Sensing Reflectance at rrs412_5
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS442_5
Long_name	Remote Sensing Reflectance at rrs442_5
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Remote Sensing Reflectance at rrs490
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Remote Sensing Reflectance at rrs510
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS560
Long_name	Remote Sensing Reflectance at rrs560
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	

Variables name in the NetCDF file	RRS620
Long_name	Remote Sensing Reflectance at rrs620
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Remote Sensing Reflectance at rrs665
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS673_75
Long_name	Remote Sensing Reflectance at rrs673_75
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS681_25
Long_name	Remote Sensing Reflectance at rrs681_25
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS708_75
Long_name	Remote Sensing Reflectance at rrs708_75
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Dataset: cmems_obs-oc_med_bgc-reflectance_my_l3-multi-1km_P1D	
Variables name in the NetCDF file	RRS412
Long_name	Multi-sensor Remote Sensing Reflectance at 412nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Multi-sensor Remote Sensing Reflectance at 443nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	

Variables name in the NetCDF file	RRS490
Long_name	Multi-sensor Remote Sensing Reflectance at 490nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Multi-sensor Remote Sensing Reflectance at 510nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS555
Long_name	Multi-sensor Remote Sensing Reflectance at 555nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS670
Long_name	Multi-sensor Remote Sensing Reflectance at 670nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Dataset: cmems_obs-oc_med_bgc-plankton_my_l3-olci-300m_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD1],[RD2],[RD3],[RD6])
Dataset: cmems_obs-oc_med_bgc-plankton_my_l3-multi-1km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD1],[RD2],[RD3],[RD6])
Variables name in the NetCDF file	CRYPTO
Long_name	Multi-sensor Cryptophyte Chlorophyll a concentration

Standard_name:	mass_concentration_of_cryptophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation ([RD7])
Variables name in the NetCDF file	DIATO
Long_name	Multi-sensor Diatom Chlorophyll a concentration
Standard_name:	mass_concentration_of_diatoms_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation ([RD7])
Variables name in the NetCDF file	DINO
Long_name	Multi-sensor Dinophyte Chlorophyll a concentration
Standard_name:	mass_concentration_of_dinophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation ([RD7])
Variables name in the NetCDF file	GREEN
Long_name	Multi-sensor Green algae and Prochlorophyte Chlorophyll a concentration
Standard_name:	mass_concentration_of_greenalgae_and_prochlorophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation ([RD7])
Variables name in the NetCDF file	HAPTO
Long_name	Multi-sensor Haptophyte Chlorophyll a concentration
Standard_name:	mass_concentration_of_haptophytes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation ([RD7])
Variables name in the NetCDF file	PROKAR

Long_name	Multi-sensor Prokaryote Chlorophyll a concentration
Standard_name:	mass_concentration_of_prokaryotes_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	The development of the algorithms applied for the PFT retrieval in this product is based on a different methodology than that used for the PFT estimate provided in other products (GLO and ATL). For more details see the relevant documentation ([RD7]).
Variables name in the NetCDF file	MICRO
Long_name	Multi-sensor Microphytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_microphytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	MICRO consists of DIATO and DINO
Variables name in the NetCDF file	NANO
Long_name	Multi-sensor Nanophytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_nanophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	NANO consists of CRYPTO and HAPTO
Variables name in the NetCDF file	PICO
Long_name	Multi-sensor Picophytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_picophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	PICO consists of GREEN and PROKAR. In the ultra-oligotrophic condition HAPTO were divided between NANO and PICO following the adjustment of [RD15]
Dataset: cmems_obs-oc_med_bgc-transp_my_l3-olci-300m_P1D	
Variables name in the NetCDF file	KD490
Long_name	OLCI Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	
Dataset: cmems_obs-oc_med_bgc-transp_my_l3-multi-1km_P1D	
Variables name in the NetCDF file	KD490
Long_name	Multi-sensor Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	

Dataset: cmems_obs-oc_med_bgc-optics_my_l3-multi-1km_P1D	
Variables name in the NetCDF file	ADG443
Long_name	Absorption due to gelbstoff and detrital material at 443 nm, QAA algorithm
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_dissolved_organic_matter_and_non_algal_particles
Units	m^-1
Additional Information:	
Variables name in the NetCDF file	APH443
Long_name	Absorption due to phytoplankton at 443 nm, QAA algorithm
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_phytoplankton
Units	m^-1
Additional Information:	
Variables name in the NetCDF file	BBP443
Long_name	Particulate back-scattering coefficient at 443 nm, QAA algorithm
Standard_name:	volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water_due_to_particles
Units	m^-1
Additional Information:	

Table 31 : list of the datasets and variable names and unit for the OCEANCOLOUR_MED_BGC_L3_NRT_009_143 product

II.3.6.4 OCEANCOLOUR_MED_BGC_L4_MY_009_144

OCEANCOLOUR_MED_BGC_L4_MY_009_144	
Dataset: cmems_obs-oc_med_bgc-plankton_my_l4-olci-300m_P1M	
Variables name in the NetCDF file	CHL
Long_name	OLCI Monthly Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD1],[RD2],[RD3],[RD6])
Dataset: cmems_obs-oc_med_bgc-plankton_my_l4-multi-1km_P1M	
Variables name in the NetCDF file	CHL
Long_name	Monthly Multi-sensor and multi water-type Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3

Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD1],[RD2],[RD3],[RD6])
Dataset: cmems_obs-oc_med_bgc-plankton_my_l4-gapfree-multi-1km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Multi-sensor, multi water-type, interpolated Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD1],[RD2],[RD3],[RD6]). The interpolated gap-free Chl concentration is estimated by means of a modified version of the DINEOF algorithm by GOS ([RD4],[RD10]).
Dataset: cmems_obs-oc_med_bgc-plankton_my_l4_climatology-1km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Climatology - Mean values of Chlorophyl Concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD1],[RD2],[RD3],[RD6]). These climatology maps have been created using the data falling into a moving temporal window of ± 5 days
Dataset: cmems_obs-oc_med_bgc-pp_my_l4-multi-4km_P1D	
Variables name in the NetCDF file	PP
Long_name	Multi-sensor Integrated Primary Production
Standard_name:	primary_productivity_of_biomass_expressed_as_carbon
Units	g m^-2 day^-1
Additional Information:	
Dataset: cmems_obs-oc_med_bgc-pp_my_l4-multi-4km_P1M	
Variables name in the NetCDF file	PP
Long_name	Monthly Multi-sensor Integrated Primary Production
Standard_name:	primary_productivity_of_biomass_expressed_as_carbon
Units	g m^-2 day^-1
Additional Information:	

Table 32 : list of the datasets and variable names and unit for the OCEANCOLOUR_MED_BGC_L4_MY_009_144 product

II.3.7 Black Sea

II.3.7.1 OCEANCOLOUR_BLK_BGC_L3_NRT_009_151

OCEANCOLOUR_BLK_BGC_L3_NRT_009_151	
Dataset: **cmems_obs-oc_blk_bgc-reflectance_nrt_l3-olci-300m_P1D**	
Variables name in the NetCDF file	RRS400
Long_name	Remote Sensing Reflectance at rrs400
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS412_5
Long_name	Remote Sensing Reflectance at rrs412_5
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS442_5
Long_name	Remote Sensing Reflectance at rrs442_5
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Remote Sensing Reflectance at rrs490
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Remote Sensing Reflectance at rrs510
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS560
Long_name	Remote Sensing Reflectance at rrs560
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS620

Long_name	Remote Sensing Reflectance at rrs620
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Remote Sensing Reflectance at rrs665
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS673_75
Long_name	Remote Sensing Reflectance at rrs673_75
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS681_25
Long_name	Remote Sensing Reflectance at rrs681_25
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS708_75
Long_name	Remote Sensing Reflectance at rrs708_75
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Dataset: cmems_obs-oc_blk_bgc-reflectance_nrt_l3-multi-1km_P1D	
Variables name in the NetCDF file	RRS412
Long_name	Multi-sensor Remote Sensing Reflectance at 412nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Multi-sensor Remote Sensing Reflectance at 443nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS490

Long_name	Multi-sensor Remote Sensing Reflectance at 490nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Multi-sensor Remote Sensing Reflectance at 510nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS555
Long_name	Multi-sensor Remote Sensing Reflectance at 555nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS670
Long_name	Multi-sensor Remote Sensing Reflectance at 670nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Dataset: cmems_obs-oc_blk_bgc-plankton_nrt_l3-olci-300m_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD8],[RD9])
Dataset: cmems_obs-oc_blk_bgc-plankton_nrt_l3-multi-1km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD8],[RD9])
Variables name in the NetCDF file	MICRO
Long_name	Multi-sensor Microphytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_microphytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3

Additional Information:	MICRO consist of Diatoms (DIATO) and Dinophytes or Dinoflagellates (DINO)
Variables name in the NetCDF file	NANO
Long_name	Multi-sensor Nanophytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_nanophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	NANO consists of Cryptophytes (CRYPTO), Haptophytes (HAPTO) and half of "Green algae & Prochlorophytes" group (GREEN)
Variables name in the NetCDF file	PICO
Long_name	Multi-sensor Picophytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_picophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	PICO consists of half "Green algae & Prochlorophytes" (GREEN) and Prokaryotes (PROKAR)
Dataset: cmems_obs-oc_blk_bgc-transp_nrt_l3-olci-300m_P1D	
Variables name in the NetCDF file	KD490
Long_name	OLCI Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	
Dataset: cmems_obs-oc_blk_bgc-transp_nrt_l3-multi-1km_P1D	
Variables name in the NetCDF file	KD490
Long_name	Multi-sensor Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	
Dataset: cmems_obs-oc_blk_bgc-optics_nrt_l3-multi-1km_P1D	
Variables name in the NetCDF file	ADG443
Long_name	Absorption due to gelbstoff and detrital material at 443 nm, QAA algorithm
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_dissolved_organic_matter_and_non_algal_particles
Units	m^-1
Additional Information:	
Variables name in the NetCDF file	APH443
Long_name	Absorption due to phytoplankton at 443 nm, QAA algorithm

Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_phytoplankton
Units	m^-1
Additional Information:	
Variables name in the NetCDF file	BBP443
Long_name	Particulate back-scattering coefficient at 443 nm, QAA algorithm
Standard_name:	volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water_due_to_particles
Units	m^-1
Additional Information:	

Table 33 : list of the datasets and variable names and unit for the OCEANCOLOUR_BLK_BGC_L3_NRT_009_151 product

II.3.7.2 OCEANCOLOUR_BLK_BGC_L4_NRT_009_152

OCEANCOLOUR_BLK_BGC_L4_NRT_009_152	
Dataset: cmems_obs-oc_blk_bgc-plankton_nrt_l4-olci-300m_P1M	
Variables name in the NetCDF file	CHL
Long_name	OLCI Monthly Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD8],[RD9])
Dataset: cmems_obs-oc_blk_bgc-plankton_nrt_l4-multi-1km_P1M	
Variables name in the NetCDF file	CHL
Long_name	Monthly Multi-sensor and multi water-type Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD8],[RD9])
Dataset: cmems_obs-oc_blk_bgc-plankton_nrt_l4-gapfree-multi-1km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Multi-sensor, multi water-type, interpolated Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD8],[RD9]). The interpolated gap-free Chl concentration is estimated by means of a modified version of the DINEOF algorithm by GOS ([RD4],[RD10]).
Dataset:	

cmems_obs-oc_blk_bgc-transp_nrt_l4-olci-300m_P1M	
Variables name in the NetCDF file	KD490
Long_name	OLCI Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	
Dataset: cmems_obs-oc_blk_bgc-transp_nrt_l4-multi-1km_P1M	
Variables name in the NetCDF file	KD490
Long_name	Monthly Multi-sensor Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	
Dataset: cmems_obs-oc_blk_bgc-pp_my_l4-multi-4km_P1D	
Variables name in the NetCDF file	PP
Long_name	Multi-sensor Integrated Primary Production
Standard_name:	primary_productivity_of_biomass_expressed_as_carbon
Units	g m^-2 day^-1
Additional Information:	
Dataset: cmems_obs-oc_blk_bgc-pp_my_l4-multi-4km_P1M	
Variables name in the NetCDF file	PP
Long_name	Monthly Multi-sensor Integrated Primary Production
Standard_name:	primary_productivity_of_biomass_expressed_as_carbon
Units	g m^-2 day^-1
Additional Information:	

Table 34 : list of the datasets and variable names and unit for the OCEANCOLOUR_BLK_BGC_L4_NRT_009_152 product

II.3.7.3 OCEANCOLOUR_BLK_BGC_L3_MY_009_153

OCEANCOLOUR_BLK_BGC_L3_NRT_009_153	
Dataset: cmems_obs-oc_blk_bgc-reflectance_my_l3-olci-300m_P1D	
Variables name in the NetCDF file	RRS400
Long_name	Remote Sensing Reflectance at rrs400
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS412_5
Long_name	Remote Sensing Reflectance at rrs412_5

Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS442_5
Long_name	Remote Sensing Reflectance at rrs442_5
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Remote Sensing Reflectance at rrs490
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Remote Sensing Reflectance at rrs510
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS560
Long_name	Remote Sensing Reflectance at rrs560
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS620
Long_name	Remote Sensing Reflectance at rrs620
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS665
Long_name	Remote Sensing Reflectance at rrs665
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS673_75
Long_name	Remote Sensing Reflectance at rrs673_75
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air

Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS681_25
Long_name	Remote Sensing Reflectance at rrs681_25
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS708_75
Long_name	Remote Sensing Reflectance at rrs708_75
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Dataset: cmems_obs-oc_blk_bgc-reflectance_my_l3-multi-1km_P1D	
Variables name in the NetCDF file	RRS412
Long_name	Multi-sensor Remote Sensing Reflectance at 412nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS443
Long_name	Multi-sensor Remote Sensing Reflectance at 443nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS490
Long_name	Multi-sensor Remote Sensing Reflectance at 490nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS510
Long_name	Multi-sensor Remote Sensing Reflectance at 510nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS555
Long_name	Multi-sensor Remote Sensing Reflectance at 555nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air

Units	sr^-1
Additional Information:	
Variables name in the NetCDF file	RRS670
Long_name	Multi-sensor Remote Sensing Reflectance at 670nm
Standard_name:	surface_ratio_of_upwelling_radiance_emerging_from_sea_water_to_downwelling_radiative_flux_in_air
Units	sr^-1
Additional Information:	
Dataset: cmems_obs-oc_blk_bgc-plankton_my_l3-olci-300m_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD8],[RD9])
Dataset: cmems_obs-oc_blk_bgc-plankton_my_l3-multi-1km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD8],[RD9])
Variables name in the NetCDF file	MICRO
Long_name	Multi-sensor Microphytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_microphytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	MICRO consist of Diatoms (DIATO) and Dinophytes or Dinoflagellates (DINO)
Variables name in the NetCDF file	NANO
Long_name	Multi-sensor Nanophytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_nanophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3
Additional Information:	NANO consists of Cryptophytes (CRYPTO), Haptophytes (HAPTO) and half of “Green algae & Prochlorophytes” group (GREEN)
Variables name in the NetCDF file	PICO
Long_name	Multi-sensor Picophytoplankton Chlorophyll a concentration
Standard_name:	mass_concentration_of_picophytoplankton_expressed_as_chlorophyll_in_sea_water
Units	milligram m^-3

Additional Information:	PICO consists of half "Green algae & Prochlorophytes" (GREEN) and Prokaryotes (PROKAR)
Dataset: cmems_obs-oc_blk_bgc-transp_my_l3-olci-300m_P1D	
Variables name in the NetCDF file	KD490
Long_name	OLCI Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	
Dataset: cmems_obs-oc_blk_bgc-transp_my_l3-multi-1km_P1D	
Variables name in the NetCDF file	KD490
Long_name	Multi-sensor Diffuse Attenuation Coefficient at 490nm
Standard_name:	volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
Units	m^-1
Additional Information:	
Dataset: cmems_obs-oc_blk_bgc-optics_my_l3-multi-1km_P1D	
Variables name in the NetCDF file	ADG443
Long_name	Absorption due to gelbstoff and detrital material at 443 nm, QAA algorithm
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_dissolved_organic_matter_and_non_algal_particles
Units	m^-1
Additional Information:	
Variables name in the NetCDF file	APH443
Long_name	Absorption due to phytoplankton at 443 nm, QAA algorithm
Standard_name:	volume_absorption_coefficient_of_radiative_flux_in_sea_water_due_to_phytoplankton
Units	m^-1
Additional Information:	
Variables name in the NetCDF file	BBP443
Long_name	Particulate back-scattering coefficient at 443 nm, QAA algorithm
Standard_name:	volume_backwards_scattering_coefficient_of_radiative_flux_in_sea_water_due_to_particles
Units	m^-1
Additional Information:	

Table 35 : list of the datasets and variable names and unit for the OCEANCOLOUR_BLK_BGC_L3_NRT_009_153 product

II.3.7.4 OCEANCOLOUR_BLK_BGC_L4_MY_009_154

OCEANCOLOUR_BLK_BGC_L4_MY_009_154

Dataset: cmems_obs-oc_blk_bgc-plankton_my_l4-olci-300m_P1M	
Variables name in the NetCDF file	CHL
Long_name	OLCI Monthly Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD8],[RD9])
Dataset: cmems_obs-oc_blk_bgc-plankton_my_l4-multi-1km_P1M	
Variables name in the NetCDF file	CHL
Long_name	Monthly Multi-sensor and multi water-type Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD8],[RD9])
Dataset: cmems_obs-oc_blk_bgc-plankton_my_l4-gapfree-multi-1km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Multi-sensor, multi water-type, interpolated Chlorophyll a concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD8],[RD9]). The interpolated gap-free Chl concentration is estimated by means of a modified version of the DINEOF algorithm by GOS ([RD4],[RD10]).
Dataset: cmems_obs-oc_blk_bgc-plankton_my_l4-multi-climatology-1km_P1D	
Variables name in the NetCDF file	CHL
Long_name	Climatology - Mean values of Chlorophyl Concentration
Standard_name:	mass_concentration_of_chlorophyll_a_in_sea_water
Units	milligram m^-3
Additional Information:	The chlorophyll estimates are obtained using two different algorithms for open ocean (case 1) and coastal (case 2) water types. ([RD8],[RD9]). These climatology maps have been created using the data falling into a moving temporal window of ± 5 days
Dataset: cmems_obs-oc_blk_bgc-pp_my_l4-multi-4km_P1D	
Variables name in the NetCDF file	PP
Long_name	Multi-sensor Integrated Primary Production
Standard_name:	primary_productivity_of_biomass_expressed_as_carbon
Units	g m^-2 day^-1

Additional Information:	
Dataset: cmems_obs-oc_blk_bgc-pp_my_l4-multi-4km_P1M	
Variables name in the NetCDF file	PP
Long_name	Monthly Multi-sensor Integrated Primary Production
Standard_name:	primary_productivity_of_biomass_expressed_as_carbon
Units	g m^-2 day^-1
Additional Information:	

Table 36 : list of the datasets and variable names and unit for the OCEANCOLOUR_BLK_BGC_L4_MY_009_154 product

II.4 Additional Variables

Depending on provider and production mode the products can include additional information such as sensor mask, quality index, errors (uncertainty) or flag quality in addition to the geophysical variable.

II.4.1 Mediterranean Sea, Black Sea and Baltic Sea Sensor additional masks

The NetCDF files of the products over the Mediterranean, Black Sea and Baltic Sea also contain an extra variable (SENSORMASK), which gives each pixel a value as a function of the sensor. For example, in the MY multi sensor product each SENSORMASK pixel is the sum of all available sensor values: SeaWiFS=1; MODIS AQUA=2; MERIS=4; VIIRS NPP=8; OLCI Sentinel-3A=16; VIIRS JPSS1=32; OLCI Sentinel-3B=64 (i.e if a pixel is observed by SeaWiFS and MERIS then SENSORMASK = 5). Moreover, for the chlorophyll product over the Mediterranean Sea, the NetCDF files contain also the water-type mask variable (WTM), which has the same dimension as the chlorophyll data, and each pixel could be a value between 0 (for case 2 waters) and 1 (for case 1 waters). For the chlorophyll product of the Baltic Sea, the NetCDF files contain also the cyanobacteria summer bloom mask variable (CYANOBLOOM), which has the same dimension as the chlorophyll data and is the sum of the values for two bloom types (subsurface =1; surface=2) thus each pixel could be a value between 0 (no summer bloom detected) and 3 (both subsurface and surface summer blooms detected).

II.4.2 Mediterranean Sea, Black Sea, Baltic Sea, Atlantic and Arctic Quality Index

This section applies only to L3 MED, BLK, BAL, ATL, ARC NRT datasets that are in the following products:

- OCEANCOLOUR_ARC_BGC_L3_NRT_009_121
- OCEANCOLOUR_BAL_BGC_L3_NRT_009_131
- OCEANCOLOUR_BAL_BGC_L3_MY_009_133
- OCEANCOLOUR_MED_BGC_L3_NRT_009_141
- OCEANCOLOUR_MED_BGC_L3_MY_009_143
- OCEANCOLOUR_BLK_BGC_L3_NRT_009_151

- OCEANCOLOUR_BLK_BGC_L3_MY_009_153

For each geophysical variable (eg. Chl, Kd, rrs, aph (not ARC), adg (not ARC), bbp) included in the products listed above a data quality information is provided in the netCDF files as an additional variable. The daily Quality Index is calculated on a pixel-by-pixel over valid observations as:

$$QI = \frac{\text{CurrentDataPixel - ClimatologyDataPixel}}{\text{STDDataPixel}}$$

with ClimatologyDataPixel and STDDataPixel being the 1km daily average and standard deviation climatology, respectively. The daily reference climatology maps have been created using independent SeaWiFS data falling into a moving temporal window of ±5 days. Single QI fields are available in the NetCDF files and can be viewed via WMS. For the current day data pixel to be considered good, QI is expected to be within 2 STD's.

II.4.3 Global Ocean and Atlantic Quality Flag and Uncertainties (Copernicus-GlobColour products)

This section applies only to GLO and ATL products (Copernicus-GlobColour products provided by ACRI-ST).

For each physical parameter (e.q. CHL, BBP…), an uncertainty variable (e.g. CHL_uncertainty) is provided. This can be a theoretical computation using LUT, variable value and observation conditions (e.g. zenith angles) or the output of the merging model. The error bar is stored in % packed into a 2 bytes integers using a scale factor of 0.01 (this kind of data packing is standard in netCDF). The biggest error bar possible in this format is 327.67, so if a computed error bar is greater than 327.67 then it is set to 327.67.

In addition, a flag parameter is provided for each netCDF file. It allows to identify land/water pixels, and for gapfree products to identify interpolated pixels. The information are stored on one byte using the CF flag encoding convention (see)

II.5 Upstream Data Unavailability

Due to the unavailability of upstream data, the following products some files can be missing for the dates listed below:

- OCEANCOLOUR_GLO_BGC_L3_MY_009_103
- OCEANCOLOUR_GLO_BGC_L3_MY_009_107
- OCEANCOLOUR_ATL_BGC_L3_MY_009_113
- OCEANCOLOUR_ARC_BGC_L3_MY_009_123
- OCEANCOLOUR_BAL_BGC_L3_MY_009_133

- OCEANCOLOUR_MED_BGC_L3_MY_009_143
- OCEANCOLOUR_BLK_BGC_L3_MY_009_153

19970905	19980710	19990125	20001117	2001118	20020730
19970907	19981117	19990126			20020731
19970908	19981118	19991117			20020801
19970911	19981119	19991118			20020802
19970912	19981120				20020803
19970913	19981217				20020804
19970914					20020805
19970917					20020918
19971013					
19971014					
19971015					
19971016					
19971017					
19971018					
19971214					
19971215					

For OCEANCOLOUR_GLO_BGC_L3_MY_009_103, concerning the PFT (part of dataset cmems_obs-oc_glo_bgc-plankton_my_l3-multi-4km_P1D) the timeseries is available from 2002-07-03 to present. Some additional dates are missing linked to missing upstream (see table below). For technical limitations (linked to the tools proposed by Copernicus Marine), PFT variables are also available in the NetCdf files provided by Copernicus Marine Service, but the files contain only “no data” for the missing products.

20020703	20030221	20040609	20060407	20070925	20101022	20110404
20020909	20030222		20060908	20070926	20101023	
20020910	20030223		20060909		20101024	20120325
20020915	20030316		20061129		20101025	
20020916	20030317		20061213		20101028	
20020924	20030318		20061214		20101029	
20021107	20030519		20061215		20101030	
20021119	20030905				20101031	
					20101101	
					20101102	
					20101103	
					20101104	

III FILE FORMAT

The format of Copernicus Marine products is NetCDF.

To know more about the NetCDF format, please follow this link: What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

In the Copernicus Marine framework, data is organized and split between "Multi Year" (MY) and "Near Real Time" (NRT) Products and Datasets.

Data can be downloaded via two different methods/protocols: Subsetter and FTP download.

The nomenclature of the downloaded files differs on the basis of the chosen download mechanism **Subsetter** or **FTP** (aka AFTP – "Authenticated-FTP", sometimes still referred as MFTP – "MyOcean FTP" for backwards compatibility, a FTP interface connected with the CMEMS Central Authentication System) service.

IV.1 Nomenclature of files when downloaded through the Copernicus Marine Web Portal Subset Service

The Subsetter Interface/Service is provided by MOTU, a Software/Interface that connects the Web Portal to the Dissemination Unit's catalogue. The details on how to access the data are provided by the Copernicus Service Desk.

The user will be asked to login using the supplied username and password. Then the user can access the data using the subsetter web interface or using a command line procedure (scripts) for automating the download (the CMEMS Service desk can provide information on how to build these scripts).

The OC files nomenclature has been harmonized in CMEMS: all Ocean Colour TAC PUs produce the files using the same nomenclature. This nomenclature is described in section IV.2. However, the subsetter does not preserve the name of the OC original files (see below), therefore, the file nomenclature differs according to the mechanism used to download them: subsetter or mftp.

The access to Ocean Colour data is provided by the two CMEMS Dissemination Units, one provides access to NRT data the other to MY data. The CMEMS Central Web Portal provides a single point of access to the whole catalog and download interfaces: https://marine.copernicus.eu/ and more specifically https://data.marine.copernicus.eu/products. The Subsetter interface requires authentication.

The filename obtained downloading a product through the subsetter interface is given by:

{CMEMS_Dataset_ID}_{random_number}.nc

Where {CMEMS_Dataset_ID} has been harmonized following the CMEMS guidelines:

{origin}_{group}-{pc}_{area}_{thematic}-{variable}_{type}_{level}-{sensor}-{spatial resolution}_{temporal resolution}

Where:

origin could be **cmems** or **c3s**;

group is **obs** (observations) ;

pc is **oc** (ocean colour) ;

area is the region code (see Table 1) ;

thematic is **bcg** (biogeochemical) ;

variable can be :

- **reflectance**
- **plancton**
- **transp**
- **optics**

type could be :

- **nrt** for "Real Time" time data
- **myint** for "Interim" data (about 1 month after the acquisition date)
- **my** for "Multi-Years" consistent time series;

level can be either **l3** or **l4** ;

sensor can be **multi** for multi sensors merged product (daily or monthly), **olci** for Sentinel-3 sensors merged product (daily or monthly), **gapfree-multi** for interpolated products and **multi-climatology** for climatology data ;

spatial resolution can be **4km**, **1km** or **300m** ;

temporal resolution could be P1D for daily product or P1M for monthly product

An example of the filename derived from daily plankton L3 multi sensor near real time product covering the Mediterranean Sea is:

cmems_obs-oc_med_bgc-plankton_nrt_l3-multi-1km_P1D_34728293.nc

Another example of the filename derived from monthly reflectance L4 Sentinel-3 OLCI sensor multi year product covering the Global Ocean is:

cmems_obs-oc_glo_bgc-reflectance_my_l4-olci-4km_P1M_89334213.nc

IV.2 Nomenclature of original files

The use of Copernicus Marine-FTP interface is typically recommended for users who:

1. want to download Copernicus Marine data regularly and automatically,
2. want to download a large amount of data (e.g. Re-analysis).
3. want to download data provided exactly by the single file

Once logged (same user/password provided by CMEMS Service Desk), users will find the following folder structure:

/Core/{Product ID}/{Dataset ID}/{temporal split}

Where "temporal split" depends on the product can be:

- "year/month/" (format "yyyy/mm/") for daily data
- "year/" (format "yyyy/") for monthly or climatological data

Browsing the desired product, users will reach the final folder containing the files. File names follow the rule:

{valid date}_{origin}_{group}-{pc}_{area}_{thematic}-{variable}_{type}_{level}-{sensor}-{spatial resolution}_{temporal resolution}.nc

The syntax follow similar rules applied to DatasetIDs, with few exceptions. In particular:

valid date yyyymmdd is the date which data refer to. For monthly or climatology data this field should be firstdate_lastdate (yyymmdd-yyyymmdd) ;

origin could be **cmems** or **c3s**;

group is **obs** (observations) ;

pc is **oc** (ocean colour) ;

area is the region code (see Table 1) ;

thematic is **bcg** (biogeochemical) ;

variable can be :

- **reflectance**
- **plancton**
- **transp**
- **optic**

type could be :

- **nrt** for "Real Time" data
- **myint** for "Interim" data (about 1 month after the acquisition date)
- **my** for "Multi-Years" consistent time series;

level can be either **l3** or **l4** ;

sensor can be **multi** for multi sensors merged product (daily or monthly), **olci** for Sentinel-3 sensors merged product (daily or monthly), **gapfree-multi** for interpolated products and **multi-climatology** for climatology data ;

spatial resolution can be **4km**, **1km** or **300m** ;

temporal resolution could be P1D for daily product or P1M for monthly product

An example of the filename derived from daily plankton L3 multi sensor near real time product covering the Mediterranean Sea for February the 13th 2022 is:

20220213_cmems_obs-oc_med_bgc-plankton_nrt_l3-multi-1km_P1D.nc

An example of the filename derived from monthly plankton L4 olci sensor, interim product, covering the Global Ocean for September 2021 is:

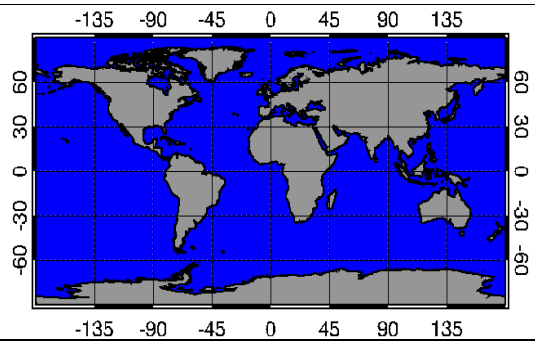
20210901-20210930_cmems_obs-oc_glo_bgc-plankton_myint_l4-olci-4km_P1M.nc

An example of the filename derived from climatology of plankton, covering the Mediterranean Sea for April the 17th is:

19980417-20100417_cmems_obs-oc_med_bgc-plankton_my_l4-multi-climatology-1km_P1D.nc

IV.3 Domain Coverage

Table below provides a graphical overview of the domain coverage for each product.

Product	Domain
Copernicus-GlobColour: OCEANCOLOUR_GLO_BGC_L3_NRT_009_101 OCEANCOLOUR_GLO_BGC_L4_NRT_009_102 OCEANCOLOUR_GLO_BGC_L3_MY_009_103 OCEANCOLOUR_GLO_BGC_L4_MY_009_104 **OC-CCI:** OCEANCOLOUR_GLO_BGC_L3_MY_009_107 OCEANCOLOUR_GLO_BGC_L4_MY_009_108	

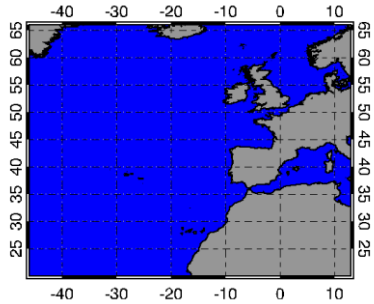
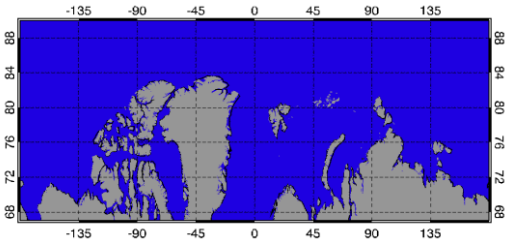
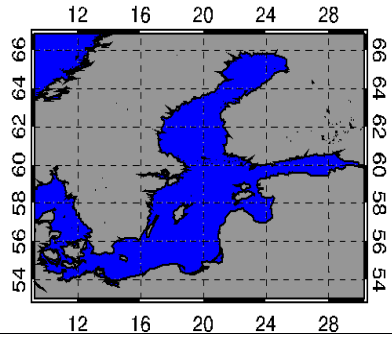
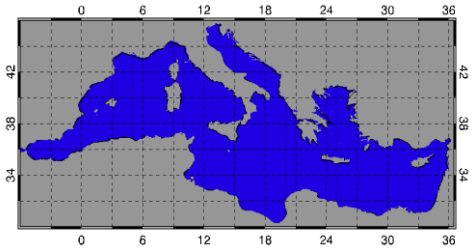
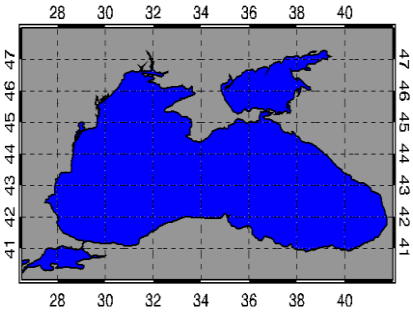
OCEANCOLOUR_ATL_BGC_L3_NRT_009_111 OCEANCOLOUR_ATL_BGC_L3_MY_009_113 OCEANCOLOUR_ATL_BGC_L4_NRT_009_116 OCEANCOLOUR_ATL_BGC_L4_MY_009_118	
OCEANCOLOUR_ARC_BGC_L3_NRT_009_121 OCEANCOLOUR_ARC_BGC_L4_NRT_009_122 OCEANCOLOUR_ARC_BGC_L3_MY_009_123 OCEANCOLOUR_ARC_BGC_L4_MY_009_124	
OCEANCOLOUR_BAL_BGC_L3_NRT_009_131 OCEANCOLOUR_BAL_BGC_L4_NRT_009_132 OCEANCOLOUR_BAL_BGC_L3_MY_009_133 OCEANCOLOUR_BAL_BGC_L4_MY_009_134	
OCEANCOLOUR_MED_BGC_L3_NRT_009_141 OCEANCOLOUR_MED_BGC_L4_NRT_009_142 OCEANCOLOUR_MED_BGC_L3_MY_009_143 OCEANCOLOUR_MED_BGC_L4_MY_009_144	
OCEANCOLOUR_BLK_BGC_L3_NRT_009_151 OCEANCOLOUR_BLK_BGC_L4_NRT_009_152 OCEANCOLOUR_BLK_BGC_L3_MY_009_153 OCEANCOLOUR_BLK_BGC_L4_MY_009_154	

Table 37 : OCTAC products domain coverage

IV.4 Structure of files

Examples of the header of output NetCDF files are inserted in annex, for each dataset.

V DIFFERENCE BETWEEN COPERNICUS-GLOBCOLOUR AND OC-CCI IN GLO MY PRODUCTS

The Copernicus-GlobColour and OC-CCI are the main initiatives in the world providing a long term series of level 3 Observation Ocean Colour products at global level based on a multi sensor approach with a 4km resolution. While both datasets aim to provide high-quality merged datasets, Copernicus-GlobColour supports continuity with the NRT production (and thus the unavoidable risk of inconsistencies in the time series due to recalibrations or processing changes in its upstream data, until a full reprocessing is done on a yearly basis to restore consistency). On the other hand, OC-CCI targets "climate quality" consistency with minimal inter-sensor bias, with the unavoidable cost of being unable to provide this consistency up to the current date (e.g. with NRT). In V5, the CCI algorithms are applied to delayed-time NRT data for the regions, reducing the difference to around a month, though these should not be considered climate grade. These two datasets therefore provide different and complementary characteristics that are useful to different users – a service offering an NRT stream may opt for the GC MY in order to be able to identify anomalies in yesterday's NRT, while a study investigating delayed-time NRT, longer subtle changes over time or deriving a historical measure for later use may opt for OC-CCI or Copernicus-GlobColour. The differences between the two initiatives are detailed in the table below.

Copernicus-GlobColour	**OC-CCI**
Project / Objective	
The Copernicus-GlobColour project started in 2005 as an ESA Data User Element (DUE) project to provide a continuous data set of merged L3 Ocean Colour products. **Since 2008**, Copernicus-GlobColour has been continuously serving **more than 600 users worldwide** with the Copernicus-GlobColour data set made available through the Hermes website. Google scholar returns **more than 400 references** to GlobColour. In 2014, a major reprocessing widely extended to a new set of products was performed thanks to financial support from the European Union Framework Program 7 under grant n°282723 **(OSS2015)** and FEDER grant n°8 562 - 40 567 (MCGS). This effort has	The ESA Climate Change Initiative is a three stage programme aiming to produce a definitive set of multi-sensor merged datasets at "climate quality" (meaning, archival grade / suitable for long term climate trend monitoring, with a complete and fully internally consistent time series) for a number of Essential Climate Variables, one of which is Ocean Colour. With the first phase of the programme beginning in 2010, OC-CCI produced the first public release (v1.0) in Q1 2013, the second phase called for annual releases (v2.0-v4.0), which include time series updates and significant targeted enhancements. The OC-CCI

been continued with a new release in April 2016 and April 2017. Beginning of 2018 the NASA R2018.0 processing has been used to reprocess the time-series. The Copernicus-GlobColour primary data set is part of the core data set of the Group on Earth Observations System of Systems http://www.earthobservations.org/geoss.php under reference: http://geossregistries.info/geonetwork/srv/en/metadata.show?uuid=urn:geoss:csr:resource:urn:uuid:4e33fd81-d5cc-dc40-b645-ab961447d9d8	programme is nearing completion of phase 3 and runs from 2019-2023. All data are freely available from https://climate.esa.int/en/projects/ocean-colour/data/ . Code for the majority of the system is also available under free licenses (excluding some portions outside the legal authority of the project) and is due to be published to github. The initial phases of OC-CCI closed in 2019 and work ws progressed in the OC-CCI+ project, providing further developments and enhancements to the multi-sensor processing chain. The OC-CCI developments are now used in the Copernicus Climate Change Service to deliver authoritative, quality-assured information on Ocean Colour.
Upstreams	
The products are based on the official level 2 data provided by agencies (NASA, NOAA and ESA). The time series is from 1997 (and still on-going) and includes SeaWifs, MERIS, MODIS, VIIRS-SNPP, VIIRS-JPSS1 (NOOA-20), VIIRS-JPSS2 (NOAA21), OLCI-S3A and OLCI-S3B.	The products are based on the best available level 2 data, as determined by two round robin comparisons made before any release – one comparing available atmospheric correction algorithms (or precomputed datasets). The criteria, methodology and results for the round robins are quantitative in the main (with a few qualitative elements) and are openly published. The v6 dataset was based on the NASA level 2 processor for SeaWIFS, POLYMER is used to process MERIS (4th reprocessing), VIIRS, OLCI-3A and OLCI-3B. This reprocessing included the updates from NASA's R2018 reprocessing of SeaWIFs.
Products overview	

GC products are available through two datasets: • one at Global Level (300m, 4km, 25 and 100km of resolution) and • the other one at European level [46W-42E 19N-72N] (1 km of resolution). The 300 meters (*) Global product corresponds to a Chlorophyll and a Reflectance product relying at present on OLCI-S3A & S3B and providing data along the coast (about 200km from the coastline). More than 40 parameters are available for **single sensor** or for **merged sensors** and for different temporal resolution: daily, 8-days, monthly, climatology. Each parameter is provided with associated information **(flags and uncertainties)** at pixelwise**.** The uncertainties computation has been published and validated ([RD17]). **The CMEMS catalogue**, a subset of the Copernicus-GlobColour products are disseminated. It concerns mainly the merged Global products at 4km: • **CHL, PP, PFT, KD, BBP, CDM, ZSD, SPM and RRS** (daily, and monthly) In addition, CHL, KD and ZSD **daily "cloud free" interpolated products** are delivered at 4km at global Level and 1km over Atlantic for CHL**.** The full time-serie (OLCI single sensor) is also delivered for the period [25 April-2016 present]	As described in the previous section, a round robin comparison is made to select the best-in-class algorithms for a release, compared against an in-situ validation database; in this case, the in-water algorithms (for chlorophyll, IOPs and Kd490). The merging process in OC-CCI uses a bandshifting approach and bias correction in order to derive Rrs that are compatible and comparable to a reference sensor (currently MERIS) to ensure maximum consistency across the complete timeseries (i.e. without significant "steps" in the data due to individual sensors producing slightly different results). The OC-CCI v6 release contains Rrs and IOPs (aph, adg and bbp) at 6 wavelengths, chlorophyll and Kd490, all with per-pixel uncertainty (RMS and bias) fields directly linked to in situ data, plus supporting variables (number of observations per sensor and IOP atot) and a 14-class water type classification, resulting in 90 data variables in total. The data are available in multiple temporal resolutions (daily, 5-day, 8-day, monthly), in two projections. All are global products at 4km spatial resolution. For CMEMS, a 1km version of the regional areas is produced. A subset of the above products is currently used (primarily Rrs to support regional algorithms). The full global 4km product is available for CMEMS.
Reprocessing (MY) Time Series	

The CMEMS MY time series available is since Sep-1997 and is completed by a CMEMS NRT time series (see next section).	The v6 release processes the full time series to the end of December 2021.
Near Real Time (NRT) products	
The NRT dataset provides the continuity with the MY. **The products are disseminated in NRT mode** (one day after satellite acquisition) and refined 30 days after acquisition. The parameters/products delivered are the same one as the MY with the same resolution. The NRT products are based on the same algorithm than the MY and have the same format. **There is continuity between NRT/MY products** useful to detect anomalies.	Although OC-CCI is primarily aimed at climate-quality outputs, an NRT mode is now run operationally. The quality of NRT data is necessarily constrained by the use of imperfect calibration and ancillary data, and processing changes imposed by upstream suppliers, so it cannot be called climate quality. The delayed-time product is identical to the MY processing (though upstream data changes, such as an unexpected NASA reprocessing, would create differences between the two).
Algorithm / Method	
In the frame of CMEMS, the algorithms used are (other alternative algorithms are available in Copernicus-GlobColour): • The Chlorophyll merging approach ([RD18]) with the capability to take benefit of OLCI-S3A, S3B and new VIIRS-JPSS1 and VIIRS-JPSS2. • BBP443, CDM443 products are obtained by the merging of MERIS, MODIS/AQUA VIIRS and SeaWiFS based on a semi-analytic model called GSM ([RD17]). The method uses an advanced retrieval based on fitting an in-water bio-optical model to the merged set of observed normalised water-leaving radiances. • The KD490 and ZSD products ([RD19]) are derived from the CHL product ([RD18]) • The SPM product is obtained using the Gohin algorithm ([RD20]).	The first parts of the OC-CCI chain produce best-in-class level 2 data per sensor, which are then binned to a global sinusoidal grid. In the second stage, these sensor data are bandshifted to the reference sensor wavelengths, then per-pixel corrected to remove bias versus this reference sensor (based on a temporally-smoothed and spatially-filled 60-day moving climatology that captures temporal and regional variation). The bias-corrected sensor data can then be simply merged. The third stage operates on the merged and bias-corrected Rrs data, computing the best-in-class in-water/derived products and a water-type classification (based on the methodology published in Moore et al (2009), but with a tuned set of water classes that best characterise the merged global data). In conjunction with an in situ database

• The PP product is based on Antoine and Morel Algorithm [RD21][RD22] • The PFT and PFS are based on [RD23][RD24][RD25] • The level-4 daily interpolated product involves multiple sensors, multiple chl-a algorithms and an interpolation scheme. The initial release ([RD16][RD26]) has been improved in 2018.	(submitted for publication) combining the majority of available validation data, per-variable and per-class uncertainty estimates can be computed, and are then applied per-pixel based on the water-type membership of each pixel (e.g. a pixel containing some percentage of coastal water types will reflect a higher uncertainty than one purely composed of pure open-ocean water). The output products are then provided additionally in a geographic projection, in a number of temporal variants and convenience subsets. For CMEMS, an additional processing at 1km is made for the regional areas, which use the corrected and merged Rrs as a high-quality input to the regional-specific algorithms.

(*) See section II.2.1

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Quality Information Document : https://documentation.marine.copernicus.eu/QUID/CMEMS-OC-QUID-009-101to104-111-113-116-118.pdf

VI ANNEX

An example of the NetCDF metadata of the daily multi-sensor plankton dataset for Mediterranean Sea file is given below:

```
netcdf 20220101_ cmems_obs-oc_med_bgc-plankton_nrt_l3-multi-1km_P1D.nc {

dimensions:
    time = 1 ;
    lat = 1580 ;
    lon = 3308 ;
variables:
    int time(time) ;
        time:long_name = "reference time" ;
        time:standard_name = "time" ;
        time:axis = "T" ;
        time:calendar = "Gregorian" ;
        time:units = "seconds since 1981-01-01 00:00:00" ;
    float lat(lat) ;
        lat:long_name = "latitude" ;
        lat:standard_name = "latitude" ;
```

```
        lat:axis = "Y" ;
        lat:units = "degrees_north" ;
    float lon(lon) ;
        lon:long_name = "longitude" ;
        lon:standard_name = "longitude" ;
        lon:axis = "X" ;
        lon:units = "degrees_east" ;
    float CHL(time, lat, lon) ;
        CHL:long_name = "Multi-sensor and multi water-type Chlorophyll a concentration" ;
        CHL:standard_name = "mass_concentration_of_chlorophyll_a_in_sea_water" ;
        CHL:type = "surface" ;
        CHL:units = "milligram m^-3" ;
        CHL:missing_value = -999.f ;
        CHL:_FillValue = -999.f ;
        CHL:valid_min = 0.01f ;
        CHL:valid_max = 100.f ;
    float WTM(time, lat, lon) ;
        WTM:long_name = "Water Type Mask for Chlorophyll a concentration" ;
        WTM:comment = "WTM = [ 1(Case1) | 0(Case2) | between the two for mixture]" ;
        WTM:type = "surface" ;
        WTM:units = "1" ;
        WTM:missing_value = -999.f ;
        WTM:_FillValue = -999.f ;
        WTM:valid_min = 0.f ;
        WTM:valid_max = 1.f ;
    int SENSORMASK(time, lat, lon) ;
        SENSORMASK:long_name = "Sensor Mask for Chlorophyll a concentration" ;
        SENSORMASK:comment = "SeaWiFS=1; MODIS AQUA=2; MERIS=4;VIIRS NPP=8; OLCI
Sentinel-3A=16; VIIRS JPSS1=32. Each SENSORMASK pixel is the sum of all available sensor
values. For example, if a pixel is observed by SeaWiFS and MERIS then SENSORMASK = 5." ;
        SENSORMASK:type = "surface" ;
        SENSORMASK:units = "1" ;
        SENSORMASK:missing_value = -999 ;
        SENSORMASK:_FillValue = -999 ;
        SENSORMASK:valid_min = 1 ;
        SENSORMASK:valid_max = 31 ;
    float QI_CHL(time, lat, lon) ;
        QI_CHL:long_name = "Quality Index for Chlorophyll a concentration" ;
        QI_CHL:comment = "QI=(DailyData-
ClimatologyMedianData)/ClimatologyStandardDeviation" ;
        QI_CHL:type = "surface" ;
        QI_CHL:units = "1" ;
```

```
        QI_CHL:missing_value = -999.f ;
        QI_CHL:_FillValue = -999.f ;
        QI_CHL:valid_min = -5.f ;
        QI_CHL:valid_max = 5.f ;
// global attributes:
        :references = "1) Volpe, G., Colella, S., Brando, V., Forneris, V., La Padula, F., Di Cicco, A., Sammartino, M., Bracaglia, M., Artuso, F., and Santoleri, R. (2019). The Mediterranean Ocean Colour Level 3 Operational Multi-Sensor Processing, Ocean Sci., 15, 127–146, https://doi.org/10.5194/os-15-127-2019. - 2) Berthon, J. F., and G. Zibordi (2004), Bio-optical relationships for the northern Adriatic Sea, International Journal of Remote Sensing, 25, 7-8, 1527-1532." ;
        :product_level = "L3" ;
        :comment = "r=log10(max([rrs443,rrs490,rrs510])/rrs555) ; chlcase1=10^[0.4567-4.2484r+2.0171r^2+2.6202r^3-2.8978r^4] ; r=log10(rrs490/rrs555) ; chlcase2=10.^[0.091-2.620r-1.148r^2-4.949r^3]" ;
        :start_date = "2022-01-01" ;
        :stop_date = "2022-01-01" ;
        :creation_date = "Fri Jan 21 2022" ;
        :creation_time = "07:01:18 UTC" ;
        :westernmost_longitude = -6.f ;
        :easternmost_longitude = 36.5f ;
        :southernmost_latitude = 30.f ;
        :northernmost_latitude = 46.f ;
        :grid_resolution = "  1 Km" ;
        :contact = "technical@gos.artov.isac.cnr.it" ;
        :netcdf_version = "v4" ;
        :grid_mapping = "Equirectangular" ;
        :software_name = "GOS Processing chain" ;
        :history = " " ;
        :source = "surface observation" ;
        :start_time = "08:00:00 UTC" ;
        :stop_time = "14:00:00 UTC" ;
        :file_quality_index = 0s ;
        :citation = " " ;
        :distribution_statement = "See CMEMS Data License" ;
        :naming_authority = "CMEMS" ;
        :cmems_production_unit = "OC-CNR-ROMA-IT" ;
        :institution = "CNR-GOS" ;
        :software_version = "v3.0" ;
        :site_name = "MED" ;
        :product_version = "v02QL" ;
        :Conventions = "CF-1.4" ;
```

:platform = "Aqua (EOS PM) satellites,JPSS-1,Sentinel-3a,Suomi-NPP" ;
:sensor_name = "MODIS,VIIRSJ,OLCI,VIIRSN" ;
:sensor = "MODIS Moderate Resolution Imaging Spectroradiometer,VIIRSJ Visible Infrared Imaging Radiometer Suite,OLCI Ocean and Land Colour Instrument,VIIRSN Visible Infrared Imaging Radiometer Suite" ;
:title = "cmems_obs-oc_med_bgc-plankton_nrt_l3-multi-1km_P1D" ;
:cmems_product_id = "OCEANCOLOUR_MED_BGC_L3_NRT_009_141" ;
:parameter_code = "PLANKTON" ;
:parameter = "Chlorophyll-a concentration and Phytoplankton Functional Types" ;